Distinction That Must Continue

Why C0/C1 Is the Only Possible Start, and the Mathematics So Far

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Abstract

We begin from one statement held whole — nothing can change; everything must change — and ask what the smallest world is that can hold it. Read from a part, the statement says *cannot hold still*; read from the whole, it says that nothing leaves and nothing is beyond. Putting only that on a finite set of states and enumerating every candidate gives a unique answer. One state is inadmissible in every reading, because its only continuation is itself. Of the sixteen relations that two states carry, exactly one survives: the swap, $a \to b$ and $b \to a$. It is a distinction (C0) and its continuation (C1) at once, and neither survives alone. It is also where every reading begins: every structure that obeys both prohibitions, read by any two-valued reader, shows both crossings of the swap, and the swap is the only structure that every such reader reads without an apparent rest. Its square is the identity, so the whole is at rest while each part moves; and because it has no fixed point, no part can read all of its own readings (Cantor, Lawvere). The start carries the reason that no reader is the whole.

The mathematics as it stands is one map read in several ways, and interaction comes first in it. A reading is the recorded outcome of an encounter that may change both participants, so the reader sits inside the system it reads. The evaluation Φ sends a state to the profile of everything its admissible encounters can show, and the address of a state is its class under that map: the classes are exactly the profiles. With continuations as the tests this is Myhill–Nerode minimality; on the paper's 24-state walk three readers give 24, 8 and 3 addresses. With a symmetry group it is the Reynolds projection, whose two bins split the 235-dimensional relation module of the series' witness 115 + 120 straight from its multiplicity vector. With declared relations it is the deficiency module $Q = V/W$, which names the missing methods without a search. Across realizations it is a commuting square, whose one-step certificate carries every word; against a declared transport it is a residual, the only place where novelty can be claimed. What is allowed and what is read are two inputs to one address: on the walk, withdrawing a move changes reachability whatever the reader, while whether it changes the addresses depends on the reader — not at all for the vertex reader, which keeps all 24, but from 6 to 4 for a reader that asks only whether the walker stands at one vertex.

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On that frame sit the keyframes of the programme — shapes counted from two prohibitions, a reader's seen-plus-hidden accounting, the two methods *continue* and *turn* with their one-bit memory, the classification of two-token loops by trace and swap, $H_1 \cong H_1(\Gamma) \oplus \mathbb{Z}/2$, the residue and its three-phase clock, prices and drives, and the realization of one game on a field and in a computer — together with a ledger of the wider corpus, forty papers read as a whole, each result placed in the frame with its status. Where a corpus result could be re-run from its own statement it was — among them the two BBP seeds, $\sec^m(2\pi/m)$ being an integer greater than 1 only for m = 6 and 8, where it is 64 and 16; the carry identity; SHA-256 run backward with its context supplied; the transport law $T_a \cap T_b = T_{\gcd(a,b)}$; and the echo of a reader. The rest is marked as reported, and the open problems are ranked. Nothing here is final. The structure is complete and its running is not, and this paper is one frame of that running.

Keywords: continuation, distinction, address, evaluation map, reader, Reynolds projection, deficiency module, transport residual, graph configuration space, exchange statistics, keyframe

MSC 2020: 05C20, 05C10, 20F36, 55R80, 05E18, 68Q45, 18D15

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1. One statement, and where it starts

1.1 One statement, read from inside

Nothing can change; everything must change. The programme reported here starts from that sentence held whole. Taken as one statement it has no consistent value, and we do not try to give it one. Split into two axioms it becomes an idea that can be computed with; held whole it is the ground the computations stand on.

The whole is complete and at rest. Any reader is a part of it, and a part is always in motion. So from every position the statement reads as change, and that change is the change of something that, taken whole, never moved. Rest, motion, time, value, order and duration are words of the part. The whole needs none of them. This paper is written from the side of the part, because that is the only side anyone writes from.

1.2 What this paper does

It does three things. Section 2 asks what the smallest world is that can hold the statement, and answers by enumeration, with the proof beside it. The answer is unique: it is C0 and C1 held as one object, the swap of two states, and every reading of every larger world begins there. Section 3 sets out the mathematics as it stands, as one evaluation map read in several ways — by continuations, by symmetries, by declared relations, across realizations and against a transport — with each instance computed, on the paper's own objects or on the smallest model that carries it. Sections 4 to 9 give the keyframes that the programme has executed or proved, now placed in that frame; Section 10 places the rest of the programme's corpus in it, separating what was re-run for this paper from what is reported; and Section 11 ranks what comes next.

The title says what the paper is about and nothing more final. An earlier version was called *The Incomplete Universe*. What that phrase meant survives in §1.3 — the structure is complete and its running is not — but a paper that reports a running should not sound like the last word on it.

1.3 Complete as structure, open as execution

Nothing here puts a hole in the world. Nothing is missing from the field and nothing falls out of it. Where a gap appears it belongs to the reader: π has no gap, a truncation of π does, and changing the reader moves

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the gap while the object stays where it was. What is open is not the structure but its execution. Three ordinary observations make the distinction exact, and the rest of the paper finds each of them again in mathematics.

- **Knowing every move does not end the game.** A player who has learned every sequence of an arcade game can still play it, repeat a route, take another, or begin again. Complete knowledge of the rules is not completion of their execution, and returning to a state is a new occurrence of it, not the old event happening twice.
- **Separation does not end potential.** A photograph outlives the moment it records. It *exists* whether or not anyone looks at it. It holds *potential* for encounters that did not exist when it was taken. And an encounter *actualizes* part of that potential — that is my house, that is my uncle, he was there — without the original moment recurring.

$$E_0 \to R, \qquad R + C_1 \to O_1, \qquad R + C_2 \to O_2$$

Here E_0 is the original event, R the retained artifact, C_1 and C_2 two later contexts, and O_1 and O_2 what each encounter produces. The artifact is independent of its source event, but not of the conditions that actualize it: no photograph produces every reading by itself, and none needs the event to happen again.

- **Knowing removes resistance without removing work.** Once a way through has been found, its organization can be kept, and a recurring situation activates it without the search being repeated. That is what a skill is, and it is what most work supplies: the repeated application of accumulated knowledge so that something else can keep going.

Put together, these say that the universe is unfinished the way a running program is unfinished, not the way a broken one is. The code can be complete while the run goes on; one run can end while another begins; and outputs keep becoming inputs. A result finished locally stays active globally. The past does not have to keep running to stay operative.

1.4 Keyframes

An animator fixes a few poses exactly and draws the frames between them to fit. A video codec stores some frames whole and rebuilds the rest from them; one paper in this programme measured exactly that trade, between storing states and storing transformations, for readers coupled to a reversible field [13]. This paper is built the same way. Each *keyframe* below is a statement that has been executed or proved, with its scope written on it.

Most of the keyframes were reached from constraint and then met, coming from the other direction, by results already in the literature: the census of strongly connected digraphs, Cantor's diagonal argument, Frobenius reciprocity, Myhill–Nerode minimality, Abrams' configuration spaces, the braid groups of graphs, quantum statistics on networks. Those collisions are the validation. A framework that only ever produces its own objects cannot be checked. One that keeps landing where established mathematics already stands is tracking something real — and when it lands there from the opposite direction, the arrival itself is information about why the result had to be true.

Two kinds of collision need keeping apart [18, §§21–23]. A *weak* collision recognizes a known theorem after the fact; it shows that the apparatus points outward, but it cannot fail once the result is known. A *strong* collision is a prediction made before the measurement that could have come out otherwise. The programme's model case is a multiplicity table that predicted a shared dimension between two families of relations at exactly the five readers inside A_4, where the scalar account read zero: direct intersection

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could have returned zero, and it returned one at all five and zero at both controls, fourteen checks out of fourteen [18, §19]. Both kinds appear below, and each keyframe says which.

We do not fill every frame between the keyframes. The frames in between are where other researchers will realize the motion in their own medium, the way baseball was realized on a field, on cardboard and on an Atari 2600. None of those is a copy of the others, and all of them are recognizably the same game, because the governing relations came first and each interface was built to carry them. Section 3.5 makes that sentence exact and Section 8.5 checks it on the paper's own object: a field and a computer play the same game when reading either one after a move gives the same address.

1.5 Status words, and the shape of the paper

Six words carry the status of every claim. **Proved**: an argument is given or cited and its hypotheses are checked. **Computed**: an exact finite computation over a stated range, reproduced by a named script (§14). **Recovered**: the statement was already published; we give the source and say from which direction we arrived. **Measured earlier**: on record in an earlier paper of the programme or in its ledger [18], and not re-run for this one. **Reported**: a result of another paper of the programme's corpus, taken from that paper as the corpus review [63] summarizes it and not re-run here; §10 lists each one with its status. The corpus ledger of §10 uses three further labels: *re-run*, for a result re-run for this paper; *in hand*, for a result that is a keyframe here; and *proposal* or *conjecture*, for an identification not yet written as a commuting square. **Open**: stated as a question, with what would settle it.

The paper follows one loop (Figure 1). Section 2 finds the only possible start, and Section 3 the one map that every later result instances, including the reading rule that an object is its address. Section 4 asks what a reader can see, Section 5 what a walker can do, and Section 6 what a loop of two walkers remembers. Section 7 separates the prices a reader can read from the drives that move things, and names the unknowns of the rates. Section 8 shows that retention — the past staying operative without running — is already present in every keyframe, and that a field and a computer play the same game exactly when their addresses agree; Section 9 closes the loop. Section 10 places the corpus, Section 11 lists the keyframes for the next stretch, Sections 12 to 14 give the ledger, the corrections and the method, and two appendices give the proofs and the scripts.

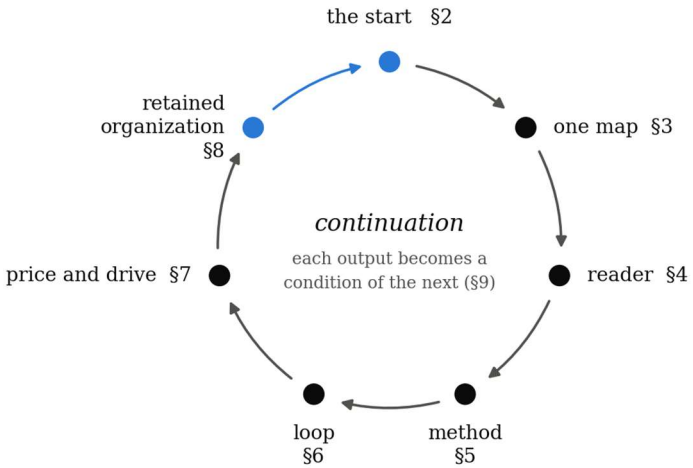


Figure 1. The arc of the paper. Each stage is a section; the last arrow returns to the first, because what a process retains becomes a condition of how it continues.

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2. The only possible start

2.1 The statement as two prohibitions

Put the statement on a finite set of states and read it twice. Read from a part, it says *cannot hold still*: every state continues, and no continuation leaves a state where it was. Read from the whole, it says that nothing leaves and nothing is beyond: every continuation stays among the states, which a relation on a set does by construction, and no state is out of reach of any other, so the graph of continuations is strongly connected — *cannot escape*. Nothing else goes in: no space, no metric, no time, no geometry, not even a number of states. The question is what can stand.

"Initial" cannot mean first in time here, since time is a word of the part (§1.1). It has to mean two things that can be checked: the smallest structure that the prohibitions admit, and the structure whose two moves appear in every two-valued reading of every structure that obeys both prohibitions. The enumeration answers both, and the answers are one object.

> **Keyframe 1. (the only possible start)** Among all structures on finitely many states that obey *cannot hold still* and *cannot escape*: (i) none has one state, whose only possible continuation is itself; (ii) exactly one has two states — the swap, $a \to b$ and $b \to a$ — out of the sixteen relations two states carry (Figure 2); (iii) above two the structures are forced and countable: 5 on three states and 83 on four. The result does not depend on which strict reading is used. Read the continuation as deterministic (a map with no fixed point and one orbit), read the whole as at rest (a bijection with no fixed point), or drop *cannot escape* altogether: one state is still inadmissible and the swap is still the only structure on two (Table 1). The weak reading, which allows rests, keeps the first of these and loses the second. *Status.* Computed exhaustively for $n \leq 4$ in five readings (`start.py`); (i) and (ii) proved in every reading (Appendix A.17). Recovered: OEIS A035512, the unlabelled strongly connected digraphs 1, 1, 5, 83, 5048, …, which differs from reading A only at $n = 1$, where the census counts one state with no continuation, which *every state continues* excludes [24].

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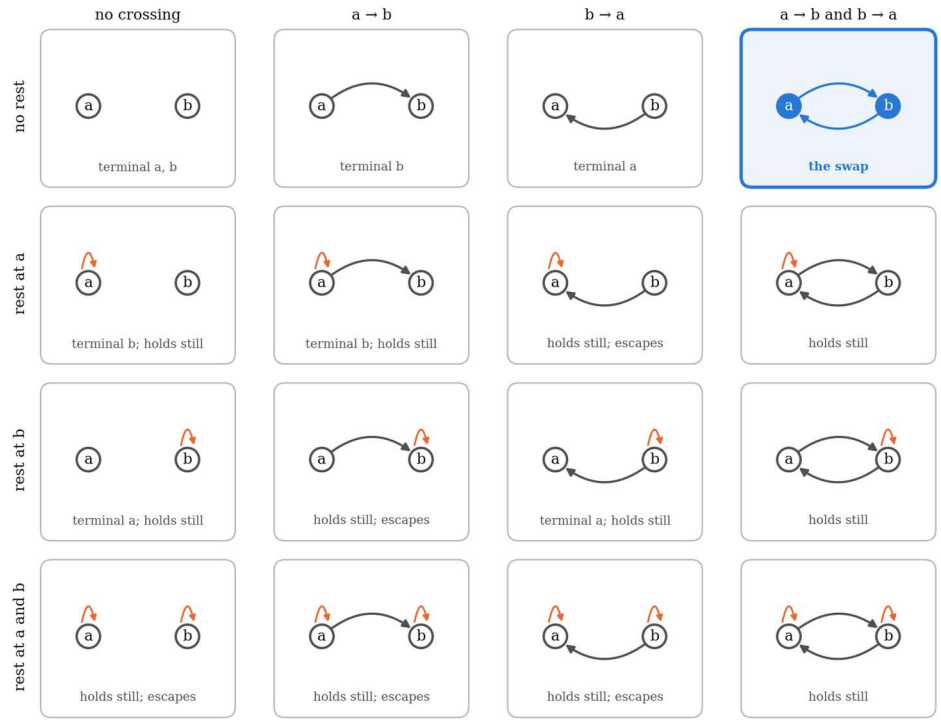


Figure 2. The sixteen relations on two states, arranged by crossings (columns) and rests (rows), each with the reason it fails. A rest is a continuation that leaves a state where it was. Only the swap survives; the three panels below it are the swap with rests added, which the weak reading D of Table 1 admits.

C0 and C1 are not two axioms here. Read the sixteen relations one at a time and the two halves of the start cannot be separated. Two states with no continuation — a distinction that does not continue — fail: both are terminal. One state that continues — a continuation with no distinction — fails: it can only hold still. A distinction with half a continuation, $a \to b$ alone, fails at b. What survives needs the distinction and its continuation at once, and it is a single object. C0, there is a distinction, and C1, every distinction continues, are two descriptions of it, the way "two states" and "the map between them" are two descriptions of one graph. At two states the first prohibition already does all the work — each state must go somewhere other than itself, and there is only one other place — so *cannot escape* comes free. It becomes an independent constraint from three states on, where dropping it leaves 7 structures instead of 5.

reading	**the prohibitions read as**	***n* = 1**	***n* = 2**	***n* = 3**	***n* = 4**
A	relation: no state continues to itself; every state continues; strongly connected	0	1	5	83
A0	reading A without *cannot escape*	0	1	7	126
B	deterministic: a map with no fixed point and one orbit	0	1	1	1
C	the whole at rest: a bijection with no fixed point	0	1	1	2
D	weak: every state *can* change — some continuation other than itself — with rests allowed; strongly connected	0	3	30	1168

Table 1. Admissible structures up to relabelling, under five readings of the two prohibitions (`start.py`; labelled counts for reading A are 0, 1, 18, 1606). Every reading kills one state. Every strict reading leaves exactly the swap on two states. Reading D answers "everything can change" rather than "everything must change", and on two states it leaves the swap with optional rests: three structures whose every change is the swap's.

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Reading D is the instructive one. It replaces *must* by *can*: a state may rest, so long as it also has somewhere else to go. One state is still dead, but two states now carry three structures — the swap, the swap with a rest at one state, the swap with rests at both — and every change any of them makes is a change of the swap. This is also where readings of C1 that are easy to confuse come apart, and there are three of them. *No terminal state* asks only that every state continue, and a state whose one continuation is itself passes it, so it admits a single state resting forever. *Not its own only continuation* — reading D — asks for somewhere else to go: it kills one state and admits rests. *No fixed point* asks that no continuation leave a state where it was, and only it keeps the start unique. The base's own wording, *everything must change*, is the strict form. What survives all five readings is what we keep: one state is never admissible, and on two states the only change there is, is the swap.

2.2 Every reading begins at the swap

Smallest is not yet initial. A structure could be smallest and still be absent from the larger ones. The swap is not absent from them, in the one sense that does not depend on how they are drawn. Take any structure that obeys both prohibitions, and any reader that sees only two values — any split of the states into two nonempty blocks.

> **Keyframe 1a. (every reading begins at the swap)** For every structure on $n \geq 2$ states that obeys both prohibitions — reading A, B or D of Table 1 — and every two-block reader, the reading contains both crossings of the swap, $0 \to 1$ and $1 \to 0$: a state in one block reaches every state in the other, and the path must cross; and it comes back. *Cannot escape* is what does this. Without it a structure can split into wholes that no crossing joins: two separate swaps on four states, read by the split between them, show no crossing at all. A move inside one block reads as rest, so the reader sees a part holding still where the field moves. The swap is the only admissible structure that every two-block reader reads exactly, with no apparent rest: on three or more states, put the two ends of any move in one block. For deterministic continuation, one cycle through all the states, an exact two-block reader exists exactly when the cycle is even. *Status.* Computed with every two-block reader over all 1 + 18 + 1,606 labelled structures of reading A and all 4 + 144 + 25,696 of reading D on two to four states, and over cycles up to length 6, with exact readers checked up to length 8 (`start.py`); proved in Appendix A.17.

So the start is the swap from two directions. From below it is the smallest admissible world and the only one of its size; from above, its two crossings are in every two-valued reading of every admissible world. Said forward, everything is built on the swap; said in reverse, everything read with two values shows the swap's two crossings, together with rests that belong to the reader. The two traversals meet at one object, and that is the sense in which C0/C1 is the only possible initial state: not first in time, but the structure whose moves every reading of every admissible world contains. The apparent rests are worth keeping in view. A coarse reader of a moving field sees stillness inside its blocks, and the stillness is the reader's. Unreadable is not absent, here in its smallest form.

2.3 The whole at rest, every part in motion

The swap s holds both readings of the statement exactly. As a part: $s(x) \neq x$ for each of its two states, and nothing holds still. As the whole: s carries the pair onto itself, $s(\{a, b\}) = \{a, b\}$, and $s \circ s$ is the identity, so two steps return every state and the whole never moved. It is also the only map on two states with no fixed point. The statement that has no consistent value when it is split — nothing can change, everything must change — has an exact model when it is held whole, and the model is the smallest thing there is.

The same property is the reason no part reads the whole. Let a reader A read itself with two values: a map f from A to 2^A, giving to each a a predicate $f(a)$ on A. Put $d(a) = s(f(a)(a))$. Then d differs from every $f(a)$ at

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a, so no f reaches every predicate. This is Cantor's argument [62], and Lawvere's diagonal theorem puts it in one line: if some A maps onto Y^A pointwise, every map of Y to itself has a fixed point; the swap has none, so nothing maps onto 2^A [40].

> **Keyframe 1b. (the start carries the reader's limit)** Because the swap has no fixed point, no map $A \to 2^A$ is onto: the diagonal $d(a) = s(f(a)(a))$ lies outside its image. Checked on every map for $|A| = 1, 2, 3$ — 2, 16 and 512 maps. With the identity in place of the swap, a map that has fixed points, the diagonal falls inside the image for 2, 12 and 296 of them, and the argument proves nothing. So a reader's incompleteness is not a further hypothesis: it comes with the start. The gap it opens is in the map f, the reader, and never in 2^A, which is all there. *Status.* Recovered: Cantor [62], Lawvere [40]; computed (`start.py`); proof in Appendix A.17.

2.4 What the start carries

The programme's other constraints apply the same requirement to one more degree of freedom at a time. **C1a**: independent distinctions remain independently continuable, so operations on independent channels commute, $[T_A, T_B] = 0$. **C2**: continuation preserves reachable history — a transition cannot erase the path that produced the present, which is where locality comes from.

The primitive object is therefore a transition relation, not a thing. A state is a temporary slice through a process, and an object is a pattern that persists because the transitions allowed to it preserve it. Persistence is resistance to collapse, not absence of change. The programme's working list of what a universe needs says the same in other words — change, retention, rules that are internal and none that are external, one common field, and an asymmetry that never balances out exactly — and several of those reappear below in computed form: retention as memory (§5.4, §8), the absence of an outside as the reader's position inside what it reads (§3.2, §4, §9), and the unbalanced asymmetry as the place where logic and the exchange sign come from (§5.5, §6.3).

The programme also carries its author's constraint chain, which applies C1 to one more degree of freedom at each rung. An executed cascade has run it rung by rung [36]; each rung re-runs a classical result, and the cascade credits each one. The numbers below were measured earlier, in that cascade, and not re-run here; and the chain numbers its rungs on its own, so its Constraint 2 is not the C2 above.

- **Constraint 2.** Every state keeps an inexhaustible supply of continuations: all 15,625 fixed-point-free maps on six states freeze their chain of images within four steps, while the successor on the natural numbers never freezes (descending chains on finite sets).
- **Law 2, location.** No matter may hold a location without the potential to be moved from it: holding one has a price that diverges, and the run returns the uncertainty floor $\sigma_x\sigma_p = 0.4999993$ in units with $\hbar = 1$. Reading Law 2 as positivity of a 2×2 Gram matrix — the same root as no-cloning — is a hypothesis of [36].
- **Transformation.** The new location inherits the law, and inheritance is conservation: momentum drifts by 2.37×10^{-14} over 200,000 steps, and by 1.20 once the well is pinned to one place (Noether).
- **Price.** A single value cannot prove change, so the price is a gradient: adding 1000 to a potential moves trajectories by 4.36×10^{-9}, and the work round a closed loop is zero for a gradient field and 2π for a curl.
- **Cost and wash.** Mass times acceleration is the cost to transform, $|ma - F| \le 2.51 \times 10^{-7}$ for $m = 1, 2, 5$; and balance reads as absence — the net force inside a $9 \times 9 \times 9$ lattice is 1.13×10^{-15}, and removing one mass makes it read $(-1, 0, 0)$ (the shell theorem).

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- **Direction, collision, field.** Direction is forced and ledgered (angular-momentum drift 1.93×10^{-13}); in a collision all change is equal, $\Delta p = \mp 1.5$ with a ledger residual of 8.88×10^{-16} (C3); and the ledger spread over a sphere gives the flux law, flux/$4\pi = 1.000000001$ for $1/r^2$ in three dimensions (Gauss).
- **Law 3.** There is no end: with finitely many states "no end" forces recurrence, a frozen orbit, which re-forces Constraint 2, so the chain closes on itself (Poincaré recurrence, credited; argued, not executed).

What is original there is the ordering — one degree of freedom per rung — not new physics. Three dimensions appear as a selection, not a derivation, and the cascade's count of correct self-predictions, 0 of 10,000, holds by construction and is not evidence [36]. The chain starts where this section did: the swap is C1 at the smallest size, and every rung keeps it.

2.5 C1 is a mobility condition

C1 has a precise form in the setting used for most of this paper. Put K tokens on a graph, at most one per vertex, and let a move slide one token along an edge to an empty vertex. On the cycle C_n the computation gives an equivalence that was not designed in.

> **Keyframe 2. (C1 is Abrams' hypothesis on the cycle)** No configuration of K tokens on C_n is stuck if and only if $n \geq K + 1$, which is exactly Abrams' condition for the discretized configuration space of C_n to have the homotopy type of the true one. The boundary row $K = n$ — every vertex occupied, nothing able to move — is precisely the violation of C1. *Status.* Computed for $n = 3, \ldots, 10$ and every K [15]; proved in Appendix A.3. Recovered: Abrams [19]; Prue–Scrimshaw [20].

C1 says that motion is possible everywhere. It does not say which motion happens, or how fast. That distinction runs through the whole paper and returns in §7 as the rate law: the constraints build the road, the rates decide the traffic, and the road already names the numbers that do it.

3. The mathematics so far: one map, read many ways

3.1 The address is the object

There is no gap between a thing and its place. Held as one statement, C0 and C1 say this: a distinction is real when some continuation tells it apart, and what no allowed continuation can tell apart is one thing, not two. So an object does not occupy an address. The object is the address — the class of everything its continuations cannot separate. "2 + 3" and "5" are two inscriptions of one address, and the number 5 is that address; a valid computation moves between inscriptions of one address and never between addresses. "2 + 3 = 8" can be written, printed or believed. Writing it is an event in the system of strings; it places nothing at a new address in arithmetic.

This is why algebra works. An unknown is not a hole waiting to be filled; it is an address the relations already fix, and solving reads it. Every extension of number that looked like filling a gap was built this way: a negative number as the class of the differences that should give it, a fraction as the class of the quotients, a real number as a cut — Dedekind's address among the rationals [33] — and a root as the variable itself read modulo its polynomial, Kronecker's construction [34]. Run for this paper: $\sqrt{2}$ read from its cut by rational comparisons alone gives 1.414213562373095, and in $\mathbb{Q}[x]/(x^2 - 2)$ the class of x squares to 2. The variable is the value. There were never any gaps to fill, only addresses not yet read.

> **Keyframe 0. (the address is the object)** Fix the continuations a system allows and what it shows. Two histories are the same state exactly when no allowed continuation separates them, and the states of the

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> smallest machine that carries the system are these classes themselves; nothing stands behind a class [28]. A proposed retained organization that merges two classes fails, its failure is a specific continuation that names the split, and splitting by failures stops at the classes [35]. On the walk of §5, with the current vertex as what is shown, "length mod 3 and total turn parity" fails at once: CC and TT end on different vertices. Adding the current vertex still fails: TTC and CCC stand on one vertex, and one continue later on opposite vertices. Splitting runs 6 → 12 → 24 classes and stops at the triples of Keyframe 8. In arithmetic the 1,110 sums of one to three digits fall into 28 classes under 121 additive contexts, one per value; "2 + 3" and "5" are one class, "8" another, and none of 1,100 valid rewriting steps leaves its class (evaluation is compositional, so this holds for every additive context; the count is a check). *Status.* Recovered: Myhill–Nerode minimality [28] and Moore's refinement [35], proof in Appendix A.1; computed for this paper (Appendix B).

Every keyframe of the paper reports an address. The swap of Keyframe 1 and the shapes above it are isomorphism classes; the carry of Keyframe 5 is the class of everything the unread positions cannot tell apart; the triple of Keyframe 8 is the class of a walk; a loop's homology class in Keyframe 12 is the pair (trace, swap) and nothing besides. An address belongs to a stated structure: among strings, "2 + 3" and "5" are two objects, and both statements hold, each in its own structure. That is two objects, not a gap between one object and its place.

The programme's address paper reached the general form first [36]. There a named law is the value that a root test returns at an address $\mathcal{A}$ = (root, class, coordinate, reader). Given only class descriptors, the loop-closure root returns nine thresholds that the literature publishes under different names — five genuine reproductions with worst relative error 1.3×10^{-11}, one correct refusal, two that hold by construction and one closed form — and one positivity inequality on a Gram matrix holds no-cloning, the cloning surface and Planck's floor. Keyframe 0 is that address with the reader slot made exact. The same paper separates three moves that this one uses throughout: a *coordinate move* inside a class is a price; a *class extension* opens coordinates that did not exist before; and a *reader change* alters what is visible without creating or destroying anything.

It carries the number 0 because every other keyframe reports an address, those of §2 included. The rest of this section shows that the programme's mathematics is one construction, of which Keyframe 0 is the first case.

3.2 One evaluation map, with the reader inside it

Interaction comes first. Two participating organizations, with state spaces X and C, meet in an encounter U_a: $X \times C \rightharpoonup X \times C$ that may change both and is defined only where the boundary admits it. A reading is the recorded outcome of the encounter, $I_a = O \circ U_a$, and neither participant holds it alone [63]. A reader is therefore specified by the encounters it can enter and the outcome it records, and its evaluation of a state is the profile of all its readouts:

$$\Phi(x) = (\, I(x, c) \,)_{c \in C}, \qquad x \sim y \iff \Phi(x) = \Phi(y), \qquad X/\!\sim \;\cong \operatorname{im} \Phi.$$

The address of x is its class, or equivalently its profile — the same object under two descriptions, which is Keyframe 0 again. When the readouts are linear the isomorphism is the first isomorphism theorem, $S/\ker \Phi \cong \operatorname{im} \Phi$, and every statement in this section is a statement about one such kernel. With continuations as the tests, $C = A^*$ and $I(x, w) = O(F_w x)$, the kernel is a right congruence — if $x \sim y$ then $F_a x \sim F_a y$ — so the classes run as a machine, $\bar{F}_a[x] = [F_a x]$ — provided the moves allowed at a state are the same at every state, or are themselves part of what is observed; otherwise two states with one profile can differ in what they may do next [63]. On the walk both moves are always allowed, and the condition holds. A probe is useful

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exactly when it splits a class that the plain observation merged; that is Moore's refinement step, one word at a time.

Two features of the definition carry weight. The reader is part of the modelled interaction: U may change the state it reads and the reader itself, so a reading is a continuation, and C1 applies to readers as it applies to everything else. And the reader has to be specified, because the space does not select its own reader: change the encounters or the outcome and the kernel can change while X does not. What a person adds is the discovery of what a specified evaluation implies. The person does not confer those implications on it.

> **Keyframe 0a. (one field, three readers)** On the 24 states of the walk of §5 — the triples of Keyframe 8 — with *continue* and *turn* as the tests, three readers give three quotients. Reading the current vertex separates all 24 states, refining 6 → 12 → 24. Reading only its sign gives 8 classes, 2 → 4 → 8, and these are exactly the states of the three-stage register with its phase counter dropped: the quotient runs, and its moves are the register's moves. Reading only its axis gives 3 classes, and no continuation ever splits them, because the axis advances by one at every step whatever the walker does: the sign never reaches this reader, a reader zero in dynamical form. The vertex reader is the pair of the other two, and its 24 addresses are exactly the pairs of a sign address and an axis address, 8 × 3: here the address factors. In every case the classes are exactly the distinct profiles, $X/\sim \cong \mathrm{im}\,\Phi$, and every move carries classes to classes. *Status.* Computed (`math_core.py`); proof in Appendix A.1.

The middle reader is the one Keyframe 17 meets from the other side. Drop the phase counter from the register and the moves still agree on all 48 pairs of state and action, but the register can no longer say where the walker is. Here the same fact is a quotient: the dynamics passes to the sign reader's 8 classes intact, and the position does not.

3.3 A reader is a projection

When the tests are the elements of a finite group H acting linearly on a space V, the reading is an average:

$$R_H = (1/|H|)\,\Sigma_{h\in H}\,\rho(h), \qquad {R_H}^2 = R_H, \qquad \mathrm{im}\,R_H = V^H, \qquad \dim V^H = (1/|H|)\,\Sigma_{h\in H}\,\mathrm{tr}\,\rho(h).$$

Its kernel is the sum of the non-trivial isotypic components — everything this reader cannot see, present in full. For a central involution z the reading splits in two, $P_\pm = (1 \pm z)/2$ with $P_+ + P_- = 1$ and $P_+P_- = 0$, and on a module with multiplicity vector m both dimensions are pairings: $\dim P_\pm V = \Sigma\, m_\lambda \dim \lambda$ over the types on which z acts as ±1. For P_+ this is Keyframe 4a's form $\langle m(V), f_H\rangle$ for the reader $\langle z\rangle$; P_- is the same pairing with the sign character, the bin 1 of Keyframe 4b.

> **Keyframe 0b. (a reader is a projection, and its zeros are addresses)** On the witness's relation module, the recorded multiplicity vector $m(A) = (15, 4, 14, 4, 24, 36)$, over types of dimensions 1, 1, 2, 2, 3, 3 on which z acts as +, −, +, −, +, −, gives 115 + 120 = 235 directly — the split that Keyframe 4b measured with an explicit signed permutation — and the same arithmetic returns the recorded splits of the commuting squares, 105 + 111, and of the rotation triangles, 10 + 10. The bins were in the multiplicity vector all along. On the walk, the central inversion z and the axis rotation g both commute with *continue* and *turn*; their averages are idempotent, of ranks 12 and 8, equal to their trace averages. A field placed on the z-odd side stays invisible to the invariant reader under every linear sequence of moves, and its square is z-even and visible at once, with its magnitudes recovered exactly and its sign not: Keyframe 4b's second-order export, on 24 states. *Status.* Computed in exact rational arithmetic (`math_core.py`); recovered: averaging, Maschke and characters [27]; proof in Appendix A.18.

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3.4 The deficiency names the missing methods

A vocabulary is a set of declared relations, and it spans a submodule W of the relation module V that the field makes available. What the vocabulary cannot say is the quotient:

$$Q = V/W, \qquad m(Q) = m(V) - m(W) = \delta, \qquad \dim Q^{H} = \langle \delta, f_{H} \rangle \text{ for every reader } H.$$

This is the structural deficiency, and it is a module, not a number. On the witness the vocabulary of commuting squares has $\delta = (2, 0, 1, 0, 2, 3)$: 19 missing dimensions, 10 on the z-even side and 9 on the z-odd side, every one of them at a computable address. The deficiency is the specification a completing family must meet — necessary, not sufficient, since four families conform to the commuting squares' deficiency and only two of them close it (§4.4) — and completing it is not addition: the union of the commuting squares and the rotation triangles is their sum minus their shared 1^{-} line, and that overlap is what makes the union an object of its own [18, §33]. At higher orders the same subtraction is Keyframe 14a: the residual at each order counts what the declared relations fail to span, dim Z − dim span $\mathcal{K}$, and each order has its own. This is the third standing aim in its general form. A missing method is a nonzero element of Q, and Q is computed, not searched for.

3.5 A square carries every word

Two realizations of one system — a field and a computer — are related by a map q from states to states, and the certificate that they are one system is local:

$$q(F_a x) = T_a(q\,x) \text{ for every } x \text{ and } a \quad \Longrightarrow \quad q(F_w x) = T_w(q\,x) \text{ for every word } w.$$

The implication is an induction on the length of w: the empty word gives the identity, and appending one action applies the one-step square once more. If the outcomes agree as well, $O'(q\,x) = O(x)$, the profiles agree, $\Phi'(q\,x) = \Phi(x)$, so addresses are carried to addresses and the two quotients are one machine (Appendix A.2). On the walk and the register, q is onto — a bijection of 24 states — and the 48 one-step checks, with the 24 checks that the register reports the walker's vertex, are the whole proof. Every word up to length 6 from every state, and the address carried by q for every such word, were checked as well, and the register read through its phase and newest bit refines $6 \to 12 \to 24$ exactly as the field does. The contract has two clauses, dynamics and address, and §8.5 shows what happens when only one of them holds.

A retained organization can be sufficient in two different senses, and the programme needs both [63]. It *recovers its source* when the source can be read back from it: the carry does that for a sum (Corpus result 2). It is *sufficient for continuation* when histories it merges behave alike under everything still to come — $r(h) = r(h')$ implies $\mathrm{Eval}_w(h) = \mathrm{Eval}_w(h')$ for every allowed continuation w — whether or not the source can be recovered. The address of Keyframe 0 is the coarsest retention that is sufficient for continuation. The sign reader's 8 classes are sufficient for continuing the sign readings and recover nothing of the walker's position; a photograph is sufficient for the encounters it supports without bringing back the event (§8.1).

3.6 Transport and residual

A claim that something is new is a claim about a quotient. Given a declared transport τ from a source relation space V_0 into a destination V_1, with $\tau(V_0) \subseteq V_1$, the residual is $V_1/\tau(V_0)$. A rise in a rank or a dimension is not novelty until the transported part has been named and taken out [41, 63]. The simplest model shows the mechanism exactly.

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Keyframe 0c. (transport has a baseline) On functions on $\mathbb{Z}/60$, let T_a be those of period a. For every pair of divisors, $T_a \cap T_b = T_{\gcd(a,b)}$ and $T_a + T_b \subseteq T_{\mathrm{lcm}(a,b)}$, so $\dim(T_a + T_b) = a + b - \gcd(a, b)$: the overlap is itself a periodic space, of period $\gcd(a, b)$. What period $\mathrm{lcm}(a, b)$ carries beyond the transported part is a residual of dimension $\mathrm{lcm}(a, b) - a - b + \gcd(a, b)$. For periods 3 and 5, 7 of the 15 dimensions of T_{15} are transported and 8 are not. *Status.* Computed for all 144 divisor pairs (`math_core.py`); proved in Appendix A.19.

The programme's BBP relation spaces have the same baseline form with measured ranks in place of periods. Subdivision transports the relations already established at the divisors, the transport baseline at $q = 15$ is $F_R(15) = R_3 + R_5 - R_1 = 9$, and the interaction residual $E_{\mathrm{int}}(15) = R_{15} - 9$ is the measurement that decides whether the frame at 15 holds only what it inherited [41]. Those ranks are reported, not re-run here, and the residual is open (§11.2).

3.7 Two waves: what is allowed, and what is read

Two different things can change in a system, and the programme keeps them apart [63]. The boundary decides which continuations are admissible, $A_B(x) = \{a : F_a(x) \text{ is admissible under } B\}$. Extraction reads a representation repeatedly, $v_k = \rho(D^k s)$, and an *echo* is a step at which the reading does not change although the representation did: $v_{k+1} = v_k$ while $D^{k+1}s \neq D^k s$. An echo belongs to the reader [48].

Keyframe 0d. (what is allowed and what is read are two inputs to one address) On all 69,888 hexadecimal strings of two to four digits, removing the leading digit leaves the integer value unchanged exactly when that digit is 0 — 4,368 echoes, 4,368 leading zeros — while removing the trailing digit is division by 16 with the remainder dropped, and it echoes only on the three all-zero strings; read as strings, neither removal ever echoes. On the walk, every pair of a boundary — both moves, *continue* alone, *turn* alone, none — and a reader — the current vertex, its sign, its axis, or the single question "is the walker at v_0?" — gives the counts of Table 2. Reachability depends on the boundary alone: 24, 3, 6 and 1 states. The address depends on both and is fixed by neither. The vertex reader keeps all 24 addresses under every nonempty boundary and falls to 6 with none; the question "at v_0?" gives 6 addresses with both moves or with *turn* alone and 4 with *continue* alone; the axis reader gives 3 under every boundary. *Status.* Computed (`math_core.py`).

allowed moves	reachable	vertex	sign	axis	at v_0
both	24	24	8	3	6
continue only	3	24	8	3	4
turn only	6	24	8	3	6
none	1	6	2	3	2

Table 2. The walk of §5 under every pair of a boundary and a reader: the states reachable from each state, which depend on the boundary alone, and the number of addresses, which depends on both (`math_core.py`).

The boundary and the reader are separate inputs, then, and the address is their joint output: what is allowed fixes what can happen, and what is allowed and what is read together fix what is told apart. A sound experiment changes one input at a time and reports every output [63]; Table 2 does that on the walk, and §11.4 asks for the same on a system where the decay can also be varied. Keyframe 6a is already a data point on the reading side: in base c a converted digit is resolved by the cell boundaries d/c of the reading, and they alone decide whether resolution is bounded.

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3.8 Four things the frame keeps apart

The frame separates four questions that a single number tends to merge, and the keyframes answer each on its own (Table 3). A capability can be available and unreadable, readable and costly, cheap and not composable, and a zero in one column says nothing about the others [50, 63].

question	what it asks	in hand
availability	which transformations are possible	C1 as mobility (Keyframe 2); *continue* and *turn* (Keyframe 8); reachability under a boundary (Keyframe 0d)
addressability	which states a reader can separate	the address (Keyframes 0, 0a); seen plus hidden (Keyframe 3); the three zeros (Keyframe 4a)
composability	which transformations combine into a method	one bit as the price of a rule (Keyframe 9); the device (Keyframe 10); unions are new objects (§3.4)
cost	what a chosen route requires	alignment buys cost, not possibility (Keyframe 6); a price is not a weight (Keyframe 16a); the carry as a retained coordinate (§10.2)

Table 3. Four quantities that the frame keeps apart, with the keyframes that measure each.

3.9 Where the mathematics stands

Table 4 is the state of the programme in one view: each line is one instance of the evaluation map, with what is read, by what reader, what the kernel merges or the quotient keeps, and what is still open. The keyframes in the last column are the evidence: those of §§2–3 are above, and Sections 4 to 9 give the rest in detail.

instance	what is read	reader	kernel or quotient	open	keyframes
start	structures on n states	two-block readers	both crossings of the swap, in every two-valued reading	—	1, 1a, 1b
continuation	states of a system	words and an outcome	the address (Nerode class)	rules that move their own readers	0, 0a, 5, 9
symmetry	a relation module	a group, by averaging	invariants; bins at computable addresses	zeros of the field: none found	3, 4, 4a, 4b, 0b
declared relations	a relation module	a vocabulary W	$Q = V/W$, the deficiency δ	which families close δ; the integral side	4c, 14a
loops	two-token configuration space	trace and swap	$H_1 \cong H_1(\Gamma) \oplus \mathbb{Z}/2$	the exchange in matter	12–15
realization	a field and a computer	a map q	one machine when profiles agree	global objects	17
transport	relation spaces	a declared transport τ	the residual $V_1/\tau(V_0)$	$E_{\text{int}}(15)$ at the BBP frame	0c
boundary and extraction	moves; representations	allowed moves; a decay D and a read ρ	reachability; echoes; addresses from both	the coupling with decay	0d, 6, 6a
prices and drives	moves of an arena	potentials	exact versus non-exact directions	the constitutive law	16, 16a

Table 4. The programme as instances of one evaluation map.

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4. Readers: what a part can see

4.1 Seen plus hidden is the whole

A reader is anything that distinguishes some states and not others: an observer, an instrument, a coarse-graining, a group of symmetries acting on configurations. The first family of reader papers in the programme took the simplest case, a reader that sees only how a permutation of N items moves the blocks of a partition [9–12].

> **Keyframe 3. (the gap is the reader's)** For every observation, the bits the reader sees plus the bits hidden in its fibre add up to $\log_2 N!$, the content of the whole. A reader with one cut can never tell a transformation from its inverse; direction becomes visible from three blocks on. And a reader's reading of orientation is a holonomy of the reader's own configuration space: the alternating group A_3 for readers that live on a line, the full S_3 in two dimensions. *Status.* Proved and computed [9, 12]; the accounting is Appendix A.4.

Nothing is lost in the accounting. What a reader does not see is held in its fibre, not removed from the world. The difference is practical: *removed* would tell us to give up, *not projected* tells us to find another address. Almost every advance in this programme has been another address to something that was already there.

4.2 The most symmetric reader is the blindest

On the arena of the series (§6) a reader is a subgroup H of the automorphism group G of the substrate, and it identifies configurations in the same H-orbit. What it can see of an irreducible representation λ is $\dim \lambda^H = \langle \chi_\lambda, \mathbb{Q}[G/H] \rangle$ — Frobenius reciprocity [27]. The trivial reader sees all six irreducible types that the witness carries; under the invariant readout the full reader sees one. The relation content that a vocabulary of declared relations leaves open is not a number but a module, and a single character, $\chi_D = \chi_A - \chi_W$, answers every reader's channel at once: fifty channels of fifteen cyclic readers from one vector of 24 entries, with no projectors and no search [4].

> **Keyframe 4. (reading the missing methods without searching)** On the witness arena the open relation content of any vocabulary is read for every reader at once from one character. Under the invariant readout the most symmetric reader is the blindest: it sees one obstruction type of six, and the other channels stay in the restricted module, unreported by that projection. Saturation at a reader is not closure: a vocabulary of 33 orbits leaves the invariant reader fully saturated while an obstruction of type 3^- remains in the module. *Status.* Computed [4, 18]; every mechanism recovered — character additivity, Maschke's theorem, Frobenius reciprocity [27].

This is the third of the programme's standing aims — find the mathematics that reads the missing methods without searching — achieved on this arena, and achieved as known mathematics. That is the outcome we wanted. The forward direction found the objects; the backward direction named them; and the names are older.

4.3 A zero is a reading, not a hole

Two results of the ledger turn the programme's oldest rule — unreadable is not absent — into arithmetic [3, 4, 18].

> **Keyframe 4a. (a reader is a functional, and the field had no gap)** Every number the ledger records at a reader H — a rank, an overlap, a loss, a residue — is a pairing $\langle m(V), f_H \rangle$ of a multiplicity vector $m(V)$

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> that does not depend on the reader with the reader's fixed-vector counts $f_H(\lambda) = \dim \lambda^H$; nine measured columns over twelve readers solve exactly as such pairings. A reader therefore changes nothing in the module; it changes which dimensions are visible, and one reader with a fixed vector in every irreducible type certifies a closure for every reader at once — on the witness the cheapest such reader has 152 coordinates against the trivial reader's 300. Zeros then come in three kinds that the arithmetic separates: *genuine absence*, $m_\lambda(V) = 0$; a *reader zero*, $m_\lambda(V) > 0$ but $\dim \lambda^H = 0$; and a *vocabulary zero*, where the declared relations do not reach the multiplicity available. The ambient module carries all six types, $m(A) = (15, 4, 14, 4, 24, 36)$. Across twelve readers, thirty-one vocabularies and every zero recorded, the first kind never occurred. *Status.* Proved (Appendix A.5); computed on the witness [3; 18, §§9–20, 34]; mechanisms recovered — Maschke, Frobenius [27].

This is the measured form of the statement that there are no gaps. Unreadable and absent look the same from one reading; the multiplicity table tells them apart without a search over readers, and on this object every apparent absence was the reader's or the vocabulary's. The same table made the strong collision of §1.4.

> **Keyframe 4b. (the invariant read is one bin; the rest is at a computable address)** For a cyclic reader of order m, restriction splits every irreducible into m frequency bins whose sizes always sum to its dimension, and the fixed-vector count used everywhere above is bin 0. Under the central involution z the relation module of the witness splits 115 + 120: the invariant read had been reporting 115 of 235 dimensions, and the other 120 sat in bin 1 at the same sample points, recoverable by one sign in the weighting. The spectator- and rotation-square families, which fall exactly one dimension short of closing together with the commuting squares in bin 0, close in bin 1. And the hidden sector is not inert. Under a dynamics that commutes with every symmetry, a field placed entirely in bin 1 stays invisible to the bin-0 reader exactly as long as the dynamics is linear; with the lowest-order nonlinearity it feeds the visible bin from the first step, and its magnitudes are recovered exactly at all 66 configurations from one visible step — only their signs need a third address. *Status.* Computed in exact rational arithmetic [18, §§35–49]; the bins are the discrete Fourier transform of the character, recovered [27].

Unreadable at bin 0 is present in full at a computable address: not somewhere in the field, but bin k, and here is k [18, §39]. That is Keyframe 0 working on data — the content never moved; the address of the read did. The wording has to stay exact. Restricting to an orbit preserves the representation; a reader that keeps only the invariant projection has discarded the rest [18, §42].

4.4 Two slots, one realization

> **Keyframe 4c. (two slots meet in a minimal realization)** Each visible number pairs a vocabulary slot with a reader slot. Over 12 reader classes and all 31 vocabularies the table of visible ranks has rank 6, the number of rational irreducibles: the slots meet in a six-dimensional interface, and neither can repair the other. A vocabulary zero — the rotation triangles carry no 2^- — is invisible at every reader; a reader zero — the Klein reader V_4 has no fixed vector in $3^\pm$ — is invisible for every vocabulary. Read as control theory, the vocabulary is the input side, the reader the output side, and the interface width is the dimension of a minimal realization: with both slots restricted the width falls strictly below the smaller of the two, to 3 and to 2 where the smaller slot is 4. That is the signature of Kalman's decomposition — a minimal realization is the part that is both reachable and observable, an intersection. *Status.* Computed [18, §§24–29]; the realization reading recovered [39].

Two consequences travel beyond the witness's numbers. A specification computed from the field — the deficiency $\delta = m(A) - m(W)$ — is necessary for a family that closes it and not sufficient: four families

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conform to the commuting squares' deficiency, and two close it [18, §31]. And combination is generative: 26 of 26 unions of families lie outside the span of their parts, because a union differs from a sum by exactly the overlap, so the redundancy is what makes a combination a new object; filling stops once the declared relations span the module [18, §33].

4.5 A cut makes a thing

A reader that groups states makes a quotient. When does a quotient make a *thing*? The adder papers answered with four clauses: a boundary; an exact compression; a transition that moves what the quotient distinguishes; and an update that reads only a bounded part of it [6, 7, 8].

> **Keyframe 5. (the carry bit is a fact about cuts)** For the n-bit adder read in any order, the minimal state that must be kept after reading the positions S depends only on S, never on the route, and factors over the maximal runs of S: $F(S) = \prod f(R)$, with $f = 1$ for the full run, 2 for a run anchored at the bottom, $L + 1$ for a top run and $2L + 1$ for an interior run of length L. The static relation residual is $b_1 = C(n, 2) + (n - 1)$. The single carried bit is a fact about bottom-anchored cuts of two-operand addition, not about binary: with m operands in radix b the carry alphabet is m, and a gap of g unread positions forgets it exactly when $m - 1 < b^g$. *Status.* Proved for all n [6, 7]; the radix and operand law computed over 220 triples (b, m, g) [8]. Recovered: Myhill–Nerode minimality [28].

Those papers paid for one lesson that is worth stating plainly: a pinned parameter is invisible to agreement. Every reading of the adder agreed while $m = 2$ and $b = 2$ were held fixed. Only changing them showed which properties belonged to the relation and which belonged to the container.

4.6 Reading without rebuilding: the address action

The programme's first standing aim is about *how* a transformation is done rather than the values it produces. Its cleanest instance is reading π at an address: the digit at position n is a function of n alone, read without building the digits before it. The Bailey–Borwein–Plouffe series [23] does this in base 16 with three operations: an address action, $x \mapsto 16^n x$, that brings depth n to the reading boundary; a fold, which turns every earlier term into a residue of bounded size; and a boundary test, which decides whether the phase sits inside one cell. Run for this paper at the addresses 0 to 11, it returns 243F6A8885A3 — the hexadecimal expansion of π, each digit read on its own.

> **Keyframe 6. (alignment buys cost, not possibility)** Reading at an address does not require the base to match the series. The hexadecimal digits of $e = \Sigma\, 1/k!$ — a series with no factor 16^{-k} anywhere — read correctly to $n = 2000$, and the decimal digits of π read directly from the base-16 series. Alignment removes the scale factor from the modulus and so changes only the cost: at $n = 1000$ the aligned read used about 47,958 modulus bits and the mismatched decimal read about 5,583,670, a factor of about 116. *Status.* Measured earlier [17]; the digit reads re-run for this paper.

The same series shows where the value of π lives. Its integral form [23] is

$$\pi = \int_0^{1/\sqrt{2}} (4\sqrt{2} - 8t^3 - 4\sqrt{2}\,t^4 - 8t^5) \,/\, (1 - t^8)\, dt,$$

with antiderivative $-2\ln(t^2 - \sqrt{2}\,t + 1) + 2\ln(1 - t^2) + 4\arctan(\sqrt{2}\,t - 1)$. The two logarithms, the radial part, vanish together at both ends of the interval; the arctangent, the angular part, carries the whole of π, rising from $-\pi$ at $t = 0$ to 0 at the fold point $1/\sqrt{2}$. The fold point satisfies $(1/\sqrt{2})^8 = 1/16$. Substituting $y = \sqrt{2}\,t$ reduces the integrand to $16(y - 1) \,/\, ((y^2 - 2)(y^2 - 2y + 2))$, whose complex poles are $1 \pm i = \sqrt{2}\cdot e^{\pm i\pi/4}$, and $(1 + i)^8 = 16$: the base and the eight-fold shell are one fact, read from one pole. The active weights (4,

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−2, −1, −1) on the residues 1, 4, 5, 6 modulo 8 sum to zero, which is what lets the radial part cancel. All of this was computed symbolically for this paper. π is read as an angle; the value is a read-off from a shape.

The same boundary test decides when a converted digit is resolved — the question the hexadecimal thread of the programme asked by measuring lookahead.

> **Keyframe 6a. (a converted digit is resolved by its boundary)** Read a hexadecimal fraction in base c. After k digits the value lies in a cell of width 16^{-k}, and its first base-c digit is fixed unless a cell boundary d/c lies strictly inside that cell. A boundary that terminates in hexadecimal is a cell end once k is large enough; one that does not terminate stays inside a cell at every k. So the worst-case number of digits needed is bounded exactly when every d/c terminates, that is for c = 2, 4, 8 and 16. For every other base it is unbounded, while the unresolved cells stay fixed in number — 2 for base 3, 8 for base 9, 14 for base 15 — so their share falls sixteenfold with each digit read. Bases 6, 10, 12 and 14 sit between: their aligned boundaries resolve and the rest never do. *Status.* Proved (Appendix A.6); computed for c = 2, …, 16 and k = 1, …, 8 (`realize.py`).

Sharing a factor with 16 is not the criterion — 6 and 12 share one and are unbounded — alignment is. This is Keyframe 6 again: alignment buys a bounded window, not the possibility of reading, since every digit in every base is still read, almost always within a digit or two. And it is resolution without exhaustion in its plainest form: a resolved digit stays resolved, and the input it came from stays available for the next digit and the next base.

5. Methods are shapes: continue or turn

5.1 The witness and its interior

Most of what follows happens on one finite object and the family that scales it. The witness X has 12 vertices and 30 edges. Six of them, of degree 7, induce an octahedron $K_{2,2,2}$, whose three antipodal pairs are its three classes. Each of the other six, of degree 3, is a leaf: it is joined to one core vertex and to both vertices of the next class. The automorphism group $G \cong C_2 \times A_4$, of order 24, acts on the core as the 24 signed cyclic permutations of three axes — the pyritohedral group. Two tokens on X give the arena: 66 configurations and 300 moves [1–5].

Paper V of the relation-layer series opened the one orbit of moves on which every invariant price vanishes and found its entire structure [5]. This *interior* consists of 12 configurations — adjacent pairs on the octahedral core — and 24 moves. The reader never exchanges the two ends of a core edge, so a state is an ordered pair (prev, cur), every interior move runs head to tail, and each state has two continuations. The two globally consistent choices, T_+ and T_-, generate a group of order 12 that acts regularly; the interior *is* its Cayley graph, a cuboctahedron; and the relators T_+^3, T_-^3 and $(T_-T_+^{-1})^2$ are exactly its 4 + 4 triangles and 6 squares.

> **Keyframe 7. (the shape is the graph of the methods)** On the witness the interior of the full reader is the Cayley graph of its two methods, and its faces are the relations among them. Shapes are methods, literally: the shape is what the methods trace, and the faces are where the methods agree. *Status.* Computed [5]; the Cayley-complex reading recovered as standard.

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5.2 Continue or turn

Under the full reader G, the memory (prev, cur) is not enough to choose a next step: the element of G that fixes a state exchanges its two futures, so no equivariant rule exists. Give the walker one more step of memory, the triple (pp, prev, cur), and the situation changes completely [18, §65].

> **Keyframe 8. (every path is continue or turn)** On the 24 triples the full reader admits exactly two deterministic rules, found by exhaustive search over all 2^{24} = 16,777,216 rule-books: **continue**, next = pp, and **turn**, next = the antipode of pp. They generate a group isomorphic to G that acts regularly and commutes with every element of the reader. Along any walk, $v[k+3] = \varepsilon_k \cdot v[k]$ (Appendix A.7), with $\varepsilon = +1$ for continue and -1 for turn, and a turn is the first difference of the method, $\varepsilon_k = m_k \cdot m_{k-1}$. A walk's endpoint is determined exactly by its length modulo 3 together with the parity of the turns at each of the three positions: 24 endpoints, checked against all 8190 words of length 1 to 12. *Status.* Computed.

The reader cannot name a method — its central element swaps T_+ and T_- — but it can tell whether the method changed, and its two rules are the two constant sequences of that one bit. Read in reverse, *continue* is the tightest loop there is, a core triangle closed in three steps, and *turn*, which changes method at every step, is the straightest line the shape has, an equator closed in six. "*x* is either this or that; every path is just continue or turn" is, on this object, a theorem.

5.3 The methods have a curvature

Joined by continue and turn, the 24 triples close up into a surface: 8 triangles, 4 hexagons and 12 squares on 48 edges, every edge on two cells, Euler characteristic 0. It is the 3.4.6.4 tiling of the plane folded by two onto a torus, and its symmetry is the classical (2, 3, 6) rotation group, recovered rather than derived [18, §65]. The same construction on the bare Platonic cores gives a sphere for the tetrahedron, a flat torus for the octahedron and a hyperbolic surface of genus 5 for the icosahedron. The sign is a count, not a coincidence: for any closed triangulated core with N triples,

$$\chi(\text{method surface}) \;=\; (\text{number of turn loops}) \;-\; N \,/\, 6,$$

so the surface is flat exactly when the turn loops have mean length 6. The turn loops are the Petrie polygons of the core, so the surface belongs to the family of maps generated by Wilson's operators [29]; the identity was checked on 15 cores [18, §66] and is proved in Appendix A.9. The witness sits on the flat case.

5.4 Memory is the price of a rule

> **Keyframe 9. (one remembered bit)** A reader admits a deterministic equivariant continuation rule exactly when no reader element fixes a state; an element that fixes a state exchanges that state's two futures. The smallest memory that lets the full reader follow a rule has 24 states — one bit more than the 12 states of the interior — and the bit need not be a vertex: the second predecessor, the last method used, or a label on the predecessor are the same G-set. *Status.* Proved by a stabiliser argument (Appendix A.8); enumerated over every transitive memory [18, §66].

This is the first place where retained organization appears as a theorem rather than an idea. The bit is exactly what has to be kept for turning to become a function. It costs one bit; without it the turn cannot be chosen at all. On the whole arena the same principle holds for every reader: one remembered step always suffices, and none is needed exactly for the readers of odd order [5].

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5.5 A device that makes a chain of logic

Fix the bits by one face and the triple becomes a three-stage register. Continue and turn act on it as

$$\text{continue: } (a, b, c) \to (b, c, a) \qquad \text{turn: } (a, b, c) \to (b, c, \neg a)$$

— a three-stage shift register closed into a ring through one controlled inverter (Figure 3). Held at continue it is a ring counter, with cycles of length 1, 1, 3 and 3; held at turn it is a twisted-ring, or Johnson, counter, with cycles of length 2 and 6 [32]. Every state accepts either instruction and produces a state, so nothing is lost and nothing dead-ends [18, §67].

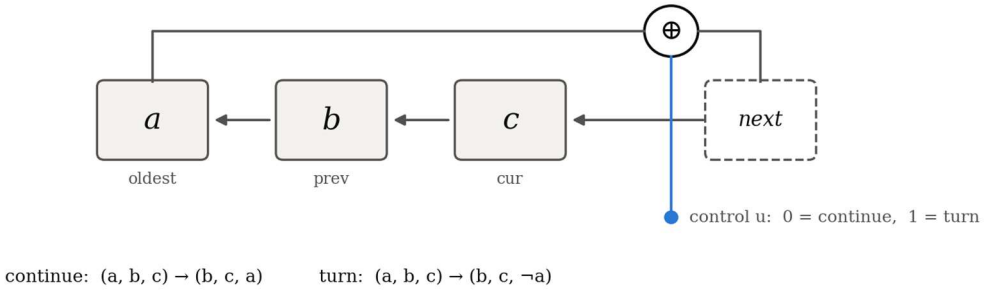


Figure 3. Continue and turn as a device. The oldest bit is fed back as the next bit, inverted when the control says turn. The register shifts; the chain never breaks.

> **Keyframe 10. (the chain is free; logic needs a broken symmetry)** Under the full reader every composition of continue and turn rotates the bits and then flips a fixed set: negation, and nothing else. As the reader is reduced, copy and exclusive-or appear, and AND — a Toffoli gate with its target on the oldest bit and its controls on the other two [31] — appears under only three readers: the order-3 rotation, a single mirror, and no symmetry at all. Under the rotation reader the 16 reversible rule-books, the invertible three-stage shift registers, generate a group of order 3·8! = 120,960 that realizes every rearrangement of the 8 bit patterns. *Status.* Computed over all 26 readers [18, §67].

So the description given to this object before any of it was computed — not a transistor, but a device that allows the physical creation of a chain of logic — is exact. Continue and turn supply a chain that never breaks; the logic that can be written on it is set by how much symmetry the reader keeps; and full reversible logic appears where the symmetry is broken. That is the programme's demand for an asymmetry that never quite balances, in executable form. Section 8.5 checks a stronger statement: with a three-step phase counter added, the register is the walk, state for state.

5.6 The shape scales

The witness is the first member of a family. Let the core be $K_{n,n,n}$: three classes of n points labelled by $\mathbb{Z}_n$, with faces the triples one per class whose labels sum to 0 or 1 modulo n, a regular triangular embedding [30]. Add one leaf for each core point, joined to it and to the whole next class; at $n = 2$ this is exactly the witness. For $n \geq 3$, give each face three new vertices in a triangle, each joined to one corner only — marks without a centre [18, §§69–71].

> **Keyframe 11. (turn, clock and one interior at every size)** For every $n \geq 3$ the automorphism group of the marked graph is the face group, of order $6n^2$. It keeps the faces, so turn is equivariant. It is transitive on the $3n^2$ core edges, so the interior is a single class. And no class rotation fixes a vertex, so the clock of §6.6 survives. With three steps of memory the rule-books number 3 at $n = 3$ and 16 at $n = 4$:

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continue, turn, and moves that leave the face. *Status.* Proved for $n \geq 3$ (Appendix A.10); computed at n = 3 and 4 (72 and 120 vertices).

The scaling also has its flat fold: on the face-structured cores $K_{n,n,n}$ alone, continue and turn close into a torus for every n computed, 2 to 7, the plane tiling folded by n, with one extra relation of order n [18, §66]. The shape that the two methods trace does not depend on the size of the object they run on.

6. Two tokens: what a loop remembers

6.1 The arena and its relation rules

With two tokens the space of situations is the arena: a configuration is a pair of occupied vertices, and a move slides one token along an edge to an empty vertex. A loop in the arena is a history. Which loops count as trivial is a declaration — a *relation rule* — and the programme measures what each declaration leaves open. On the witness the short loops fall into five families (Table 5).

family	what happens	on the witness
commuting square	the two tokens move on disjoint edges, then move back	291
spectator square	one token waits; the other goes round a 4-cycle of the graph	528
rotation square	both tokens move, round one 4-cycle of the graph	264
spectator triangle	one token waits; the other goes round a triangle of the graph	180
rotation triangle	both tokens move, round one triangle of the graph	20

Table 5. The five families of short loops in the two-token arena of the witness [1, 5]. Every one of them swaps the two tokens or not in a fixed way: rotation squares swap, the other squares do not.

Declaring the commuting squares alone gives Abrams' discretized configuration space [19]. For two tokens on a simple graph it has the homotopy type of the space of two unlabelled points moving on the graph without colliding [19, 20]: the space whose loops are the braids of two particles on a network. Its first homology is the abelian record of those braids.

6.2 Trace and swap

Every loop of two unlabelled tokens can be read in two ways. Its **trace** J is the closed path the two tokens draw on the graph together. Its **swap** ε is 1 if, following labelled tokens round the loop, they finish exchanged, and 0 if they do not. Both are well defined on homology classes, because a commuting square traces nothing and swaps nothing.

Keyframe 12. (a loop's class is its trace and its swap) Let Γ_n be the n-th member of the family, Γ_2 the witness. Then the sequence $0 \to \mathbb{Z}/2 \to H_1(D_2\Gamma_n; \mathbb{Z}) \to H_1(\Gamma_n; \mathbb{Z}) \to 0$ is exact, and (J, ε) is an isomorphism $H_1(D_2\Gamma_n; \mathbb{Z}) \cong H_1(\Gamma_n; \mathbb{Z}) \oplus \mathbb{Z}/2$. The free rank is the graph's own cycle rank, $\beta_1 = 12n^2 - 3n + 1$ for $n \geq 3$ and 19 for the witness, and the only class belonging to two tokens is the exchange z, of order exactly 2. Every exchange — at every branch point, through every three arms, in either direction — is the same element z, because each one traces nothing and swaps. *Status.* Computed in exact integer arithmetic for n = 2, …, 6 (Table 6); for every n by Theorem 4 of Harrison, Keating, Robbins and Sawicki [21], with the hypotheses checked below; the full argument is Appendix A.11. Recovered in part from Ko and Park [22].

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n	graph V / E	β_1	arena: configurations / moves / commuting squares	H_1
2	12 / 30	19	66 / 300 / 291	$\mathbb{Z}^{19} \oplus \mathbb{Z}/2$
3	72 / 171	100	2,556 / 11,970 / 13,239	$\mathbb{Z}^{100} \oplus \mathbb{Z}/2$
4	120 / 300	181	7,140 / 35,400 / 41,922	$\mathbb{Z}^{181} \oplus \mathbb{Z}/2$
5	180 / 465	286	16,110 / 82,770 / 102,330	$\mathbb{Z}^{286} \oplus \mathbb{Z}/2$
6	252 / 666	415	31,626 / 166,500 / 212,049	$\mathbb{Z}^{415} \oplus \mathbb{Z}/2$

Table 6. Integral first homology of the two-token arena under commuting squares, from a spanning-tree presentation reduced by exact unit-pivot elimination. In every case all surviving relations are multiples of twice the exchange, and one token going round each fundamental cycle while the other waits off it gives a legal loop with that trace and swap 0. The counts at n = 4, 5, 6 were also reported independently and agree exactly [18, §§74–75].

The theorem that carries this to every n states that for a 3-connected simple graph, $H_1(D_2(\Gamma)) = \mathbb{Z}^{\beta_1(\Gamma)} \oplus A$, with $A = \mathbb{Z}_2$ for non-planar graphs and $A = \mathbb{Z}$ for planar ones [21, Theorem 4]; their D_2 is exactly our arena with commuting squares. The family meets the hypotheses at every size. Γ_n is simple. It is non-planar: it contains $K_{3,3}$ for $n \geq 3$, and the witness was checked directly. It is 3-connected: deleting any two vertices leaves the core $K_{n,n,n}$, which is $2n$-connected, in one piece; every leaf keeps one of its $n + 1$ core neighbours; and every mark vertex keeps one of three internally disjoint routes to the core. No vertex cut of size two or less exists at $n = 2, \ldots, 6$, which was also checked by exhaustion.

The splitting is canonical, and it is physical. The section that sends a graph cycle to a loop is: one token goes round the cycle while the other waits anywhere off it — and where it waits does not matter, because every such loop has the same trace and swap 0. Figure 4 shows the other half, the exchange itself, as six keyframes.

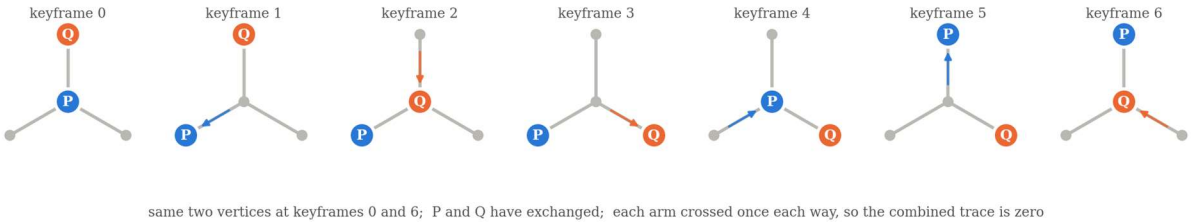


Figure 4. The exchange in six keyframes. Two tokens, P and Q, pass each other through one branch point in six moves. At keyframe 6 the same two vertices are occupied as at keyframe 0, but the tokens have exchanged; each arm is crossed once in each direction, so the combined trace is zero and the swap is one. On a planar graph this loop has infinite order. On a non-planar graph twice it bounds — on the 18 squares of the two-token surface of $K_{3,3}$ the signed sum has boundary exactly $2z$ — and it is the element of order two.

6.3 Why the swap is ±1: the exchange as physics

Among 3-connected graphs the order of the exchange is decided by planarity alone (Table 7). On a planar graph the exchange is a free direction and adds one to the rank; on a non-planar graph it has order two.

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graph	planar	β_1	H_1 of two tokens
K_4	yes	3	$\mathbb{Z}^4$
octahedron	yes	7	$\mathbb{Z}^8$
cube	yes	5	$\mathbb{Z}^6$
dodecahedron	yes	11	$\mathbb{Z}^{12}$
K_5	no	6	$\mathbb{Z}^6 \oplus \mathbb{Z}/2$
$K_{3,3}$	no	4	$\mathbb{Z}^4 \oplus \mathbb{Z}/2$
Petersen	no	6	$\mathbb{Z}^6 \oplus \mathbb{Z}/2$
$K_{3,4}$	no	6	$\mathbb{Z}^6 \oplus \mathbb{Z}/2$
Möbius–Kantor	no	9	$\mathbb{Z}^9 \oplus \mathbb{Z}/2$

Table 7. Two tokens on 3-connected graphs, computed by the same integral reduction. Planar: β_1 + 1 free directions. Non-planar: β_1 free directions and one class of order two.

Read through phases, this is quantum statistics. A quantum particle hopping on a network picks up a phase around each loop, and the abelian phases available to two indistinguishable particles — the one-dimensional unitary representations — are the characters of this H_1. On a planar 3-connected network the exchange can carry any phase — anyonic statistics. On a non-planar one it can carry only ±1: bosons or fermions [21]. Representations of the braid group of higher dimension lie outside this count. The free directions are the one-particle loops of the graph, through which each particle picks up its own Aharonov–Bohm-type phase; the exchange is the only genuinely two-particle class.

The construction of this paper started from two prohibitions and a reader that fixes the direction of three classes, and it arrived at the configuration space whose topology fixes the statistics of particles. The witness shows the step that matters. Its core octahedron alone is planar; the leaves, which are the decoration that gives the reader its direction, are exactly what make the graph non-planar and pin the exchange to ±1.

> **Keyframe 13. (the exchange is the orientation character of a surface)** The two-token complex of $K_{3,3}$ has 15 configurations, 36 moves and 18 commuting squares, and every vertex link is a single cycle — a hexagon where both tokens sit on one side, a square where they are adjacent — so the complex is a closed surface. It is non-orientable, with Euler characteristic −3, hence of genus 5. The exchange reverses orientation: its mod-2 class is Poincaré dual to the first Stiefel–Whitney class w_1, the mod-2 cup form on the surface is non-degenerate of rank 5, and $w_1(z) = 1$. *Status.* Computed; the cup-product data agree with an independent run [18, §74].

6.4 More tokens

The exchange bit does not depend on how many particles there are. With k tokens the discretized model is the configuration space once every edge is subdivided into $k - 1$ pieces [19, 20]. On that model the first homology is the same for $k = 2$, 3 and 4 on every graph tested (Table 8), and on the non-planar ones the exchange, with the other tokens parked, is still the element of order two — in agreement with the theorem that for 2-connected graphs the first homology stops changing with the number of particles [21, Theorem 5]. K_5 with four tokens reproduces the worked example of Ko and Park, $H_1(B_4K_5) \cong \mathbb{Z}^6 \oplus \mathbb{Z}_2$ [22]. Without the subdivision the discrete model can give the wrong answer — $K_{3,3}$ with three tokens and no subdivision gives $\mathbb{Z}^5$ — a reminder that an arena has to be the space it stands for.

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graph	$k = 2$	$k = 3$	$k = 4$	$k = 3$, unsubdivided
K_4 (planar)	$\mathbb{Z}^4$	$\mathbb{Z}^4$	$\mathbb{Z}^4$	$\mathbb{Z}^3$
cube (planar)	$\mathbb{Z}^6$	$\mathbb{Z}^6$	—	$\mathbb{Z}^6$
$K_{3,3}$	$\mathbb{Z}^4 \oplus \mathbb{Z}/2$	$\mathbb{Z}^4 \oplus \mathbb{Z}/2$	$\mathbb{Z}^4 \oplus \mathbb{Z}/2$	$\mathbb{Z}^5$
K_5	$\mathbb{Z}^6 \oplus \mathbb{Z}/2$	$\mathbb{Z}^6 \oplus \mathbb{Z}/2$	$\mathbb{Z}^6 \oplus \mathbb{Z}/2$	$\mathbb{Z}^6 \oplus \mathbb{Z}/2$
witness	$\mathbb{Z}^{19} \oplus \mathbb{Z}/2$	$\mathbb{Z}^{19} \oplus \mathbb{Z}/2$	—	$\mathbb{Z}^{19} \oplus \mathbb{Z}/2$

Table 8. First homology with k tokens, each edge subdivided into k – 1 pieces (k = 3: into 2; k = 4: into 3). The last column omits the subdivision.

6.5 The residue: one circulation survives every square

Declare every 4-cycle of the arena instead of the commuting squares alone. The exchange dies: a spectator square and a rotation square on the same 4-cycle of the graph have the same trace and opposite swaps, so declaring both kills z. And of the 181 free directions at $n = 4$ exactly one survives. It survives at every size.

> **Keyframe 14. (the residue)** For every n the first rational cohomology of the arena with all its 4-cycles filled is one-dimensional: H^1(square-filled arena of Γ_n; $\mathbb{Q}$) $\cong \mathbb{Q}$. It is spanned by the pull-back of an integral cocycle f on the graph: $f = +1$ on a core edge from class i to class $i + 1$; from a leaf, 3 to its own core point and 1 to each point of the next class; 0 from a mark vertex to its corner; and $f(t_a \to t_b) = f(a \to b)$ on a mark triangle. The cocycle f vanishes on every 4-cycle of the graph and is ±3 on every triangle. *Status.* Proved (Appendix A.12): every arena 4-cycle either traces nothing, and is then a commuting square, or traces one 4-cycle of the graph, and every 4-cycle of the graph is traced; by Keyframe 12 this turns the question into one about the graph with its 4-cycles filled, whose first homology is spanned by a single triangle. Computed at arena level for $n \le 4$ and at graph level for $n \le 8$ [18, §§72, 74, 75].

The residue is the forward bias around the three classes, $0 \to 1 \to 2 \to 0$. It belongs to the object rather than the reader: no group is needed to define it, and the reader acts on it only by +1. Of all the circulations the graph carries, it is the one that every square relation lets through.

6.6 The clock: the residue read modulo 3

Reduce f modulo 3. On every edge $f(u \to v) \equiv \mathrm{cl}(v) - \mathrm{cl}(u)$, where cl is the class number of a vertex and a leaf or a mark vertex takes the class of its own core point. So modulo 3 the residue is exact, and its potential is the class number. On the arena the potential is $F(\{u, v\}) = 2(\mathrm{cl}\ u + \mathrm{cl}\ v) \bmod 3$. It is single-valued — and it is not invariant, because a reader element g that rotates the classes shifts it:

$$F(g \cdot x) \;=\; F(x) + \mathrm{rot}(g) \quad (\mathrm{mod}\ 3).$$

> **Keyframe 15. (the clock is the residue at another coefficient ring)** The rational residue and the reader's order-3 clock are one integral cocycle read over $\mathbb{Q}$ and over $\mathbb{Z}/3$. The clock is the reader-relative class: the character $G \to \mathbb{Z}/3$ by which class rotations shift F. Its kernel on the witness is the reader of order 8, whose three classes on the interior are the three phases of the clock. It survives exactly when no class rotation fixes a pair of token positions. *Status.* Proved (Appendix A.13); computed at $n = 2, 3, 4$ [5; 18, §§68–71, 74].

An earlier lemma of the programme says the same in the language of periods. When every circulation of a class lies in $d\mathbb{Z}$, the class reduced modulo d is exact, so a single-valued clock always exists [16]. Nothing is missing between the two readings. Over $\mathbb{Z}/d$ the cocycle is a potential, the clock; over $\mathbb{Z}$ it is a circulation,

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the residue; and the winding that the clock does not carry is the residue itself. One object, read at two addresses, and d is the number that joins them.

6.7 One operation at every order

The relation rules of this section are one instance of a single operation — declare some differences equivalent and keep what survives — and the operation recurs, one order at a time [37]. States are compared and a reader identifies some; transitions are compared and exact cochains are set aside; routes are compared and declared squares identify some; relations are compared, and a declared cube identifies those.

> **Keyframe 14a. (the residual counts what the declared relations fail to span, at every order)** A declared cell is a cycle, so the cochains it kills are the annihilator of the span of the declared cells, and $\varepsilon_{harm} = \dim Z - \dim \text{span}(\text{declared cells})$, where Z is the cycle space at that order. Declaring nothing leaves every cycle; declaring more can only lower the residual; declaring every cycle closes the order. One order up, three pairwise disjoint moves generate a commuting cube — a relation among relations — with $d_2 d_1 = 0$ exactly, and the residual of the cubes is not fixed by the residual below: the prism with three tokens is closed at the route order and keeps one class one order up, $(\varepsilon_{harm}, \varepsilon^2_{harm}) = (0, 1)$; the bull is the reverse, (1, 0); and a third substrate keeps (1, 3). *Status.* Proved (Appendix A.14); computed on ten substrates, with the identity checked in 20 of 20 cases at the first order and 12 of 12 at the second [37].

Three consequences follow. A zero at one order says nothing about the next: a calculation can be complete at its own order and silent about the one above. A residual channel has a lifetime over added relations: on two bridged 4-cycles, adding the extra orbits of 4-cycles one at a time kills seven channels at seven different thresholds [37]. And curvature belongs to a cochain together with a relation rule: on the eight-vertex substrate of the series one cochain has zero circulation on all 89 commuting squares and circulation ½ on 24 of the 165 arena 4-cycles — flat under one rule, curved under the other — while exactness, fixed before any relation is declared, does not move [37]. This also constrains the end of the author's chain: if "all change is equal" means changes declared the same, then equality of change is a relation on changes, not a scalar recoverable from the changes themselves, and any Law 3 must name the equivalence it uses [37].

7. Price, drive, and the rate law

7.1 A static potential drives nothing

A *price* is the difference of a potential, $p = d\varphi$. Paper IV showed that an invariant price is constant on a reader's classes, so it is silent on every move inside a class; Paper V opened the moves where it is silent and found the methods of §5 there [4, 5]. Silence is not the only limit. If motion is driven by a price alone the system relaxes to detailed balance: transitions continue, but no net current flows. On the cycle arenas tested a static potential produces no current at all, and adding a potential to any other drive changes the current through the residue channel by exactly nothing [15]. The potential is not weak at driving the channel. It is invisible to it.

7.2 Drives, and what a relation rule lets through

Write a rate law on the arena, with forward and backward rates $k(u \to v)$ and $k(v \to u)$, and let $a = \log(k(u \to v)/k(v \to u))$ be its antisymmetric part. By Kolmogorov's criterion the dynamics satisfies detailed

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balance — it settles into a stationary state with no current — exactly when a is exact, $a = -\beta\, dE$ [25]. Everything else is a *drive*, measured by its circulation around loops: the cycle affinities of Schnakenberg's network theory [26]. On the witness, with rates that respect the reader, the count is the following.

reader-invariant directions of the antisymmetric part a	**on the witness**
all of them (one per orbit of moves)	22
prices: exact, the equilibrium landscapes	7
drives: not exact, the cycle affinities	15
drives with zero affinity round every commuting square	2
… and round every 4-cycle	1 — the residue
… and round every triangle as well	0

Table 9. Prices and drives on the witness [18, Part I and §75]. Zero affinity round the commuting squares is the antisymmetric half of independence: the log-ratio of each token's rates does not depend on where the other token sits. Full independence also needs the mobilities of §7.3 to factor. The drives that remain are then the one-token drives, through the trace of Keyframe 12.

So the relation rules of §6 are statements about which loops a dynamics can drive. Rate ratios that respect independence admit only one-token drives, the invariant cycles of the graph. Balance around every 4-cycle leaves one drive, the residue of Keyframe 14: a steady push forward around the three classes. Balance around every short loop leaves none.

> **Keyframe 16. (the exchange cannot be driven, only phased)** Under any drive that respects independence the exchange has zero affinity: twice the exchange is a boundary, so $\oint_{\mathrm{Z}} a = \tfrac{1}{2} \oint_{\partial \mathrm{B}} a = 0$. Real rates cannot pump a persistent swap. Only a phase sees it, through $\mathrm{Hom}(\mathbb{Z}/2, \mathrm{U}(1)) = \{\pm 1\}$. *Status.* Proved (Appendix A.15).

The three kinds of structure are now separate and each has its own instrument. Prices are equilibrium landscapes and are read by potentials. Drives are circulations and are selected by relation rules. The exchange is an integral class of order two, invisible to every real-valued rate and visible to a phase.

7.3 The rate law and its unknowns

First, where the 22 comes from, and why the textbook operator is not the answer [18, §§50–54].

> **Keyframe 16a. (the operator is not forced; a zero price is a move the reader cannot see)** On the arena, the linear operators that commute with every symmetry, act only between neighbouring configurations, are self-adjoint and conserve the total form a 22-dimensional family, one free weight for each of the 22 orbits of moves. The usual graph Laplacian is the single point where all 22 weights are equal, and nothing in those four conditions makes them equal. Requiring that a signed price be the difference of a potential cuts its freedom to 7 parameters — 8 classes of configuration less a constant — because an invariant price can only come from an invariant potential. Price and weight are different objects: a price is antisymmetric and a weight symmetric, the weights $|d\varphi|$ span 18 of the 22 directions, and the Laplacian is excluded only if a zero price forces a zero weight. A zero price falls on every move whose two ends the reader cannot tell apart — on the witness the 24 moves inside one class of configurations — for every invariant potential, and for a potential that separates all eight classes nowhere else. *Status.* Proved (Appendix A.16); computed [18, §§50–54].

So a price vanishes exactly where its own reader is blind, and the operator that would turn prices into motion is a choice among 22 weights that symmetry, locality and conservation leave open.

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What the topology does not choose is the symmetric part of the rates — the mobility $\kappa_{uv} = \kappa_{vu}$, one number per orbit of moves, 22 on the witness — and the scale β that turns a price into a rate ratio. Local detailed balance makes the split explicit:

$$k(u \to v) \;=\; \kappa_{uv} \cdot \exp(-\beta\, p_{uv} / 2), \qquad \text{so} \qquad k(u \to v) \,/\, k(v \to u) \;=\; \exp(-\beta\, p_{uv}).$$

It is a candidate closure, not a derivation: it fixes the ratio from the price and leaves the speed to κ. On a move of zero price the forward and backward rates are equal but need not be zero — a move can be unpriced and still be fast. One structural consequence is already forced. A demand for sustained winding round a residue cannot be the equilibrium energy of a closed system: on the cover where the winding becomes single-valued, such an objective is unbounded below in one direction, so a persistent demand for winding is necessarily open, driven, finite in horizon, or tied to a task [15].

None of this is a hole in the world. Every realization of an arena has its rates. The topology reads their antisymmetric part — prices and drives — and names the rest without choosing it. The unknowns are located: the symmetric part of the rate law. They are counted: 22 mobilities and one scale on the witness. And they are read by measurement: forward and backward rates, the stationary current that a drive produces, and the ratio test $\log(k(u \to v)/k(v \to u)) = -\beta(E_v - E_u)$. In any realization the variables already have their values. What the topology leaves open is only which realization the world uses, and that is read, not searched for.

8. Retained organization: the past does not have to keep running

8.1 The photograph, again

Return to the photograph of §1.3. Three things are true of it at once. It *exists* as an artifact. Its organization can *support* further interactions. And a particular encounter *selects* some of its distinctions and produces a result. It can exist without being examined; its potential does not need to be continuously activated; and its next use need not reproduce the event. The original event and the retained artifact occupy different continuations: the artifact is produced by one history and can enter another.

None of this is a picture laid over reality by a framework. The photograph exists, the recognition happens, the decision follows. NEXUS does not grant those events reality by naming them; it tries to describe what they do.

8.2 Learning moves the work

Let a system have a working state x, a retained organization m, and an incoming encounter e, and let one joint rule U update both:

$$(x_{t+1}, m_{t+1}) \;=\; U(x_t, m_t, e_t), \qquad T_m(x, e) \;=\; \mathrm{pr}_x\, U(x, m, e).$$

The joint rule never changes. The *effective* rule T_m — how this system now handles that encounter — changes whenever m does, so the same event can produce a different response before and after: $T_{m_0}(x, e) \neq T_{m_1}(x, e)$. No new law is needed for learning; the state was larger than it looked. Continuation can change the structure that determines continuation.

Before learning, the next move is selected by search. After it, a recognized situation activates a retained organization and the move is executed. The work of finding a continuation has become structure that supplies the continuation. That relocates cost rather than abolishing it, and an accounting that keeps the parts apart is

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$$C_{\text{total}}(N) = C_{\text{acquire}} + C_{\text{maintain}}(N) + N \cdot C_{\text{run}},$$

under which retention pays when acquisition lowers the recurring cost enough over N encounters. A boat coming up onto the plane is the mechanical picture: still powered, still meeting the water, but in a different regime of continuation. Cumulative culture is the same thing at the scale of generations: a person learns a method without repeating the centuries that produced it, and the method becomes a component of work its discoverers could not have reached. Retained organization keeps the full cost of the past from becoming a recurring entrance fee to the future. The compression that matters here is compression of the reconstruction process, not of bits: a large representation can save enormous work, and a small one is useless if using it means rediscovering the result.

8.3 The mathematics already does it

Every keyframe of §§4–6 is, among other things, a photograph: a retained organization that is independent of the history that produced it and sufficient for the next use (Table 10).

what is retained	what it is enough for	keyframe
one bit: the last method, or the second predecessor	choosing continue or turn as a function	9
length mod 3 and three turn parities	the endpoint of any walk	8
trace and swap	the class of any two-token loop	12
the clock modulo d	which of d sheets a history ends on	15
the reader cut, $F(S)$ states	continuing the addition from any set of read positions	5
seen bits, with the fibre held	the whole content, $\log_2 N!$	3
a walk read as three register bits and a phase	continuing the walk on a computer	17

Table 10. Retained organizations already proved or computed in the programme. Each forgets something definite: the order of moves, the lift of the clock, the content of the fibre.

In each row the past has stopped running and remains operative. Each row is an address in the sense of Keyframe 0, the class of histories that the next use cannot tell apart, so what is retained is the address itself and not a record of how it was reached. Each row also says exactly what the retained organization does not carry — the order of the moves, the non-abelian braid, the lift of the clock, the content of the fibre — and that is what makes it usable. A photograph that tried to keep everything would be the event again.

8.4 Reacting and reorganizing

Two kinds of execution need to be kept apart. *Reactive* execution propagates a change through an existing organization: an input arrives and the dependent responses follow, as in reactive programming. *Reflective* execution also changes the organization that will handle the next input. Looking at the photograph and recognizing the uncle revises what the next look will find:

$$e_t = \text{EMMIT}(r, m_t, c_t), \qquad m_{t+1} = \text{UPDATE}(m_t, e_t, c_t), \qquad e_{t+1} = \text{EMMIT}(r, m_{t+1}, c_{t+1}).$$

The artifact r has not changed; the organization m that encounters it has, so the same artifact can emit something new. EMMIT need not create information from nothing — the new relation comes from combining distinctions in the artifact with distinctions the reader already holds. Not every reaction reorganizes, but where one does, the boundary between the two kinds is measurable in any computational system, which makes it the natural first test of a retained-continuation law (§11.9).

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8.5 Baseball on the field and in the computer

A constraint structure delimits a space of continuations; actual interactions make one of them concrete; and the realization changes what is available next:

$$C \to \mathcal{P}(C) \to R \to C'.$$

Baseball shows each arrow. Its governing relations — outs, innings, bases, the conditions under which one action changes the state of the game — came before every interface that carries them. The field, the board game and the Atari 2600 cartridge each realized enough of those relations to remain baseball while changing the motions completely, and each then became an environment in which people learned, played and built further games. The cartridge was pulled into existence by rules that already existed, and the timing matters: the rules precede the interface that realizes them. Once it existed it changed what could happen next. The direction can also reverse — a physical event is sensed and encoded, a computed design drives a machine, and the result becomes input again — and neither the analog nor the digital form is terminal.

What makes the field and the cartridge the same game can now be said exactly. Let x be a configuration of either realization, F_a its own evolution under an action a — a pitch, a swing, a button press — and q the reading that returns the state of the game. A realization plays the game when reading after acting is the same as acting on the reading:

$$q(F_a(x)) = T_a(q(x)) \qquad \text{for every configuration } x \text{ and every action } a.$$

The field evolves by mechanics and the cartridge by circuit logic, and neither needs the other's matter. By Keyframe 0 the state of the game is an address — the class of configurations that no allowed continuation of play can tell apart — and when q is that class map the square commutes automatically (Appendix A.1). Two realizations therefore play the same game exactly when their addresses agree, that is, when their smallest machines for the game's observations are the same machine [28, 35]. The field and the computer are then one object read twice. Everything outside the game's contract — grass, leather, voltages — is what each realization keeps to itself, and at those addresses they are different objects.

> **Keyframe 17. (the field and the computer are one object when their addresses agree)** The paper's own object supplies a field and a computer, built independently. The field: walks on the octahedral core of the witness, each state the last three vertices, *continue* returning to the vertex two steps back and *turn* going to its antipode — built from adjacency alone. The computer: a three-stage ring register whose fed-back bit passes through a controlled inverter, with a three-step phase counter — built from bits alone. The reading q sends a walk's three vertices to the axis of the oldest and the three signs. Then q is one-to-one and onto the 24 states; $q(F_a(x)) = T_a(q(x))$ holds for all 48 pairs of state and action; all 2,046 words of length 1 to 10, from every state, give the same answer on both sides; and the computer reports the walker's vertex correctly. Drop the phase counter and the square still commutes, three states to one, but the register can no longer say where the walker stands: the bits (1, 1, 1) occur at two different vertices. *Status.* Computed (`realize.py`); the sameness of smallest machines recovered [28, 35, 39].

The register is not a picture of the walk; at the address the walk's continuations define, it is the walk. The same square appears twice more in this paper. Trace and swap (Keyframe 12) is a reading of loops: it sends every loop of two tokens to (trace, swap), keeps the homology class, and forgets the order of moves and every rate. The serial adder is a reading of arithmetic (Keyframe 5): a circuit that writes one digit at a time and carries one bit realizes 2 + 3 = 5 and then 5 + 3 = 8, because reading its output gives the sum; drop the carried bit and 5 + 3 reads as 6, wrong in three of four places [18, §77]. The square can fail in

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only two ways, and both point somewhere: the realization does not implement the rule, or the reading has merged two states that some continuation separates — and then the failing continuation names the missing distinction, exactly as in Keyframe 0.

The arrows run both ways, and they branch. Mechanics is read into a game state and the game is written into a cartridge; a design computed on a screen drives a machine, and the machine's result is read again. Each realization becomes a source in its own right — the cartridge permits replays, statistics and new games on the same rules, which the field never could — and its results become constraints elsewhere. In the terms of [36], a coordinate move inside a class is a price, while a class extension opens coordinates that did not exist before. A new capability meeting a constraint that was already there is such an extension: the method it opens is available as soon as the two can be related, before anyone builds it, and it is new for the reason the ledger measured on the witness — a union differs from the sum of its parts by exactly their overlap, so a combination carries directions that neither part had [18, §33].

Section 3.5 gives this square its general form, and the 48 one-step checks are already a proof for words of every length; the long runs check the code, not the claim. Each such correspondence needs a contract that names its states, actions and observations, the map, what the map preserves, what it forgets, and its scope, together with a falsifier — a continuation on which the two sides would differ if the correspondence were only verbal [63]. The three readings above are contracts of exactly this kind: the register with its phase counter preserves everything the walk can show and forgets nothing; the register alone preserves the dynamics and forgets the position; trace and swap preserves every homology class and forgets the order of moves and every rate. An equivalence is always relative to what it promises to keep, and naming the kernel is what lets it travel between realizations without being stretched beyond them.

9. The loop closes

The pieces now fit together. The structure is complete. The start is one object, the swap; the catalogue of admissible moves on every arena of this paper is finite and fully written; the first homology of two tokens is known for every size; the methods of a walker are two; the residue is one. The execution is not complete. Walks continue, loops are traversed again, and what a walk keeps of its past becomes part of how it handles the next step. A result finished locally stays active globally. None of this puts a hole in the field. The gap is always the reader's — the start already carries it (Keyframe 1b) — and every keyframe above was found by changing the reader — a coarser group, one more remembered step, a different relation rule, another coefficient ring — rather than by adding anything to the object. Nor is there a gap between an object and its address: each keyframe is the class of what its continuations cannot tell apart (Keyframe 0), and finding one was always reading an address the relations had already fixed. On the witness the field never showed a gap at all: every zero recorded was a reader's or a vocabulary's (Keyframe 4a), and every hidden channel had a computable address (Keyframe 4b).

We are not outside this. A game cartridge and a planet are very different realizations, but neither stands outside the constraints it runs on, and neither do we; a field and a cartridge are one game exactly where their addresses agree (Keyframe 17). The readers of this paper are subsystems of the systems they read (§3.2), and there is no exterior from which a complete reading could be taken. That is not a limitation to regret. It is the reason the running continues — a universe that one of its parts could read completely would leave its parts nothing to do — and the reason every finished result becomes a starting point. The past is retained as organization; the organization is met by new conditions; the meeting produces what did not exist before; and that becomes organization in turn.

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The optimism of this paper is specific. Starting from one statement and two prohibitions, the programme has built objects that keep landing on established mathematics and physics: the census of strongly connected digraphs, Cantor's diagonal argument, Frobenius reciprocity, Myhill–Nerode minimality, Abrams' configuration spaces, the braid groups of graphs, the exchange statistics of particles on networks. Each landing is a confirmation the construction could not have arranged for itself. The claims that were killed on the way (§13) were wrong turns caught by the instruments, and each one sharpened the road. What remains to be read is named, located, counted and measurable: the rates, with their unknowns written down, and the residuals of §11, with their baselines. With the keyframes in place, the route between them can be drawn by anyone who builds a realization that respects them.

10. The forty papers in one frame

10.1 What the corpus review found

The programme's wider corpus was read as a whole for this version through an AI review of forty papers and a revised, interaction-first synthesis of them [63]; one of the forty, named in §10.4, is not the programme's work. Every item of that review that can be checked from its own statement was re-run for this paper, and the others are marked *reported*. The review's central finding is the frame of §3. The papers fit as a typed architecture: admissibility says which moves are legal; closure selects what satisfies the constraints jointly; a reader maps configurations to observations; continuation says which distinctions stay resolvable; a retained coordinate can restore what a projection omits; and a transport comparison separates inherited structure from a genuine residual. They do not fit as one equation, and the distinctions between those parts are what let them fit. Table 11 places every item with its status, the cross-domain proposals included, and the review's ranked open problems are folded into §11.

10.2 What was re-run here

Three corpus results were re-run inside §§2–3: the two formulations of continuation in *Two Origins* [50] — typing a successor into the space, which excludes nothing, and forbidding a state to be its own only continuation, which does — are the remark of §2.1 and reading D of Table 1; the intersection law of *The Move Layer* [41] is Keyframe 0c; and the echo of *The Reader Is the Variable* [48] is Keyframe 0d. Three more were re-run for this section.

> **Corpus result 1. (exactly two seeds)** With $\theta = 2\pi/m$, $1 + i\tan\theta = \sec\theta \cdot e^{i\theta}$, so $(1 + i\tan\theta)^m = \sec^m\theta$: the base of a seed is the m-th power of a pole. It is an integer greater than 1 for exactly two values, $m = 6$, where $(1 + i\sqrt{3})^6 = 64$, and $m = 8$, where $(1 + i)^8 = 16$; the frames (64, 6) and (16, 8) of the two-angle classification are these two [42, 43]. The second is the pole $1 \pm i$ that §4.6 read from the integral of π. The reason is Galois: if $\sec^m(2\pi/m)$ is rational, every conjugate of $2\cos(2\pi/m)$ has the same absolute value, which leaves degree at most two, so $m \in \{1, 2, 3, 4, 5, 6, 8, 10, 12\}$, and checking those leaves 6 and 8. *Status.* Proved (Appendix A.20); scanned for $m \le 200$ at 80 significant digits, with the two hits exact (`corpus_rerun.py`). The scope is the corpus's own: the classification is of the positive primitive secant class, and over all relation-bearing kernel frames a blanket two-seed claim is false — the negative-base frame (−64, 12) lies outside it [46] (reported).
>
> **Corpus result 2. (a retained coordinate repairs a projection)** For any sum of two 32-bit words, $S = W + 2^{32}Q$ with W the low word and $Q \in \{0, 1\}$ the carry. The pair (W, Q) recovers S; W alone is two-to-one — at 4 bits every low word comes from two sums except the top one, which comes from one. Determinism is not injectivity. A projection is made lossless by naming the coordinate it omits and

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keeping it, which is what the carry of Keyframe 5 does at a cut [51]. *Status.* Proved: the identity is division with remainder, and the pair is one-to-one for the same reason; the 4-bit table shows the two-to-one projection in full (`corpus_rerun.py`).

Corpus result 3. (SHA-256 runs backward when its context is supplied) With the round's message word W_t and constant K_t supplied, each round inverts in closed form. The new state's words b', c', d', f', g', h' are the old a, b, c, e, f, g; then $T_2 = \Sigma_0(a) + \mathrm{Maj}(a, b, c)$, $T_1 = a' - T_2$, $d = e' - T_1$ and $h = T_1 - \Sigma_1(e) - \mathrm{Ch}(e, f, g) - K_t - W_t$, all modulo 2^{32}. The last expression is one-to-one in W_t, so with the word withheld the previous state is fixed in seven of its eight words and runs through every value in the eighth. The schedule recurrence solves for its oldest term, $W_{t-16} = W_t - \sigma_1(W_{t-2}) - W_{t-7} - \sigma_0(W_{t-15})$, so any 16 consecutive schedule words give back the message block. Local reversibility with its context is not inversion of the digest, which is a separate many-to-one projection; the programme's SHA work measures where the remaining difficulty sits [14] (§11.10). *Status.* Proved by the algebra above (Appendix A.21); checked on one message block against the standard library — all 49 windows, all 64 rounds, and the withheld word at round 0 (`corpus_rerun.py`); the round function and schedule are those of the standard [64].

10.3 The ledger of the corpus

corpus result	papers	slot in the frame	status here
two positive primitive secant seeds, $m = 6$ and 8 ($b = 64$ and 16); lifts addressed by (m, k)	[42, 43]	closure: joint constraints select	the seeds re-run and proved (Corpus result 1); the lifts reported
no blanket two-seed law over all kernel frames: the frame $(-64, 12)$	[46]	the scope of a closure	reported; a correction made in the corpus
the coefficient map of $16 - t^8$: rank 7, a one-dimensional kernel, a two-dimensional π-fibre; the coefficient-sum functional isolates the BBP numerator	[44, 45]	closure: a distinguished fibre	reported
the two angle classes alone do not force the stencil $(4, -2, -1, -1)$; it needs pole or annihilator assumptions as well	[43, 44]	closure: what selects the readout	reported; open (§11.3)
a kernel dimension depends on its coefficient field; at (16, 24) it is 3, not 5	[46]	the field is part of the reader	reported; a correction made in the corpus
incidence one forces a coordinate to vanish, two or more frees it, and a freed coordinate need not be occupied: (64, 42) is freed and unoccupied	[47]	availability is not occupancy	reported; occupancy open (§11.3)
subdivision transports relations: $T_a \cap T_b = T_{\gcd(a,b)}$, baseline $F_R(15) = 9$	[41]	transport and residual	the intersection law re-run in the periodic model (Keyframe 0c); the BBP ranks reported; the residual open (§11.2)
a value is a reading, an echo is the reader's failure, and periodicity needs a reader with state	[48]	extraction	the echo re-run (Keyframe 0d)
a probe informs exactly when it splits an observation fibre	[49]	a continuation reader	in hand: Moore's step (§3.2)

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corpus result	papers	slot in the frame	status here
typing a successor into the space excludes no map; forbidding a state to be its own only continuation does	[50]	the start	re-run as readings D and A (Table 1); that typing excludes nothing holds by construction
the carry restores the modular projection exactly; no loss holds only where a retained coordinate is named	[51]	a retained coordinate	proved (Corpus result 2)
a SHA-256 round inverts given its word; the schedule runs back from 16 words; the digest is many-to-one	[14]	a realization with its context	proved; checked on one block (Corpus result 3)
two quotients of one machine carry different invariants	[6]	two readers, two kernels	in hand (Keyframe 5)
each relational order has its own residual	[37]	declared relations	in hand (Keyframe 14a)
a residual is an invariant of the declared complex, not of the bare substrate	[53]	declared relations	reported; consistent with Keyframes 12–14
a reader-null sector is sourced at second order; 22 conserving operators, none forced	[4]	a reader as a projection	in hand (Keyframes 4b, 16a, 0b)
state-space semantics: bisimulation and predicate abstraction as reader-relative quotients	[59]	a continuation reader	reported; the form of §3.2
cycle types in double cosets through Eulerian transition systems	[60]	readers of permutations	reported
the sphere is an orbit; directed path length is a quasi-metric, symmetric under reversibility	[52]	a symmetry reader	reported; classical
the isoperimetric deficit vanishes only at the sphere; every geometric quantity is a boundary quantity	[52]	the boundary wave	the inequality classical; the universal claim open by the source's own ledger
two four-operation vocabularies: FOLD, CLOSE, PARTITION, CARRY and ADDR, TURN, SELECT, ACCUM	[54]	operations	open: no refinement map between them yet
the origin layer: distinction, boundary, shape, reader, retained organization	[61]	the start and its order	reported; §2 gives its first step by enumeration
weight distribution follows task granularity in a small network; a temporal drain	[55]	cost and resolution	measured; the drain failed in its own setup (reported)
Taylor–Green at 64^3 and Re = 400: the strain trace closes, the vorticity aligns with the intermediate eigenvector, production over dissipation falls below one	[56]	a cost ledger	measured on one finite run (reported); the uniform depletion bound open
an $n - 1$ count across SHA registers, SU(3) generators and BBP sectors	[57]	a correspondence	conjecture: no commuting square yet
a damped mode and the BBP construction; a universal transport system	[58]	a correspondence	proposal: the square of §3.5 is its test

Table 11. The corpus placed in the frame of §3. Re-run *means re-run for this paper;* in hand *means the result is a keyframe here;* reported *means taken from its paper through the review [63] and not re-run;* proposal *and* conjecture *mark identifications not yet written as a commuting square.*

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10.4 What stays out, and why

Two kinds of item stay outside the frame's claims. *The Emergence of Spacetime and Mathematical Coherence Through Recursive Harmonic Architecture* was among the forty papers reviewed, but it is not this programme's work, and nothing here uses it. And identifications across domains — between SHA-256's registers and SU(3), between a damped oscillator and the BBP series, between a four-operation vocabulary and a universal instruction set — appear in Table 11 only as proposals. They enter the frame when their square is written down: states, actions, observations, a map, what it keeps, what it forgets, and a continuation on which the two sides would differ if the identification were only verbal. That is the contract of §3.5, and the walk and the register are its worked example.

11. Where it is going

These are the keyframes for the next stretch, roughly in order of how much each would unlock [63], each with what would settle it.

11.1 The selection theorem

Every section of this paper is an instance of one conditional statement that has not yet been proved in general. Joint constraints select a structure when exactly one address survives them and weakening an operative constraint reopens the class; a retained coordinate repairs a reader exactly when the augmented reading is one-to-one on the distinctions it promises to keep; and a residual is new only over a declared transport. Section 2 is its first worked case — on two states the first prohibition does all the work and leaves one address, and weakening it from *must* to *can* reopens the class to three — and the BBP seeds (Corpus result 1), the walk and the register (Keyframe 17) and the carry (Corpus result 2) are the next three. What would settle it is a theorem with typed hypotheses that covers all four.

11.2 The interaction residual at 15

Measure R_{15} in the BBP relation frame with the transported basis recorded, form the quotient by the transported span, and report the independent new directions. Three outcomes are possible — the frame at 15 holds only what it inherited, inherited directions collide, or a direction appears that transport does not explain — and only the third is generation [41]. The number $E_{\mathrm{int}}(15) = R_{15} - 9$ cannot tell them apart by itself: it is the count of new directions minus the count of collisions, and the two can cancel. The quotient does tell them apart: its dimension counts the new directions, and the shortfall of the transported span below 9 counts the collisions. Keyframe 0c is the control, the model in which the transported span and the residual are both computed exactly.

11.3 Occupancy, and the stencil

The partner predicate forces some coordinates to vanish and frees others, and a freed coordinate is not always occupied [47]: (64, 42) is freed and unoccupied. A criterion that predicts both the occupied and the unoccupied cases, including the documented counterexample frames, is the next result on the BBP line. In the terms of Table 3 it separates availability from a fifth question, occupancy, as §3 separates availability from addressability. Beside it sits the stencil. The two angle classes of the two-angle theorem do not by themselves force the weights (4, −2, −1, −1) that §4.6 reads; a proof that they do under stated pole or annihilator assumptions, or a counterexample showing that the stencil needs a condition of its own, settles it [43, 44].

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11.4 Two waves on one system

Hold the boundary fixed and vary the decay and the read; then hold those fixed and vary the boundary; then couple them — on one finite system, measuring reachability, addresses, residuals and cost separately. Keyframe 0d crossed every boundary with every reader on the walk (Table 2) and measured extraction alone on hexadecimal strings; no decay was varied on the walk. The hexadecimal decay witness is the natural system for the full experiment: there the extraction of Keyframe 0d and the resolution boundaries of Keyframe 6a already act on the same strings, and an admissibility boundary — which digits may follow which — can be added and varied on its own.

11.5 From local certificates to global objects

Every keyframe above is a certificate of one kind: a quotient closes, a square commutes, a threshold is crossed. A local certificate can certify a local step; it does not by itself certify the global object that a physical or computational claim needs — an attracting orbit, a generated class, a completed instrument [38]. The programme's address paper found exactly this slot empty in its own registry and made the first measurement there. A delayed four-stage positive loop whose local condition predicts a cycle for $\tau > 0.1644$ shows oscillation at a fixed run length; but at fixed delay the oscillating fraction falls as the run is lengthened — from 37 to 20 of 48 histories at $\tau = 5$, and from 9 to 0 at $\tau = 2$, as the run length goes from 500 to 4000 — so the cycle is a long transient, as the theory of monotone systems requires of a positive loop [36]. The next package is therefore global: order structure for loops with a sign pattern, which the literature already supplies, and then index theory or gluing obstructions. On a finite object the global form is already visible in Keyframe 17 — a square checked on every state and every action — and the items below are the places where that form has to be reached.

11.6 Constitutive selection

Measure the mobilities and the scale on a physical realization of an arena. Test the ratio law $\log(k(u \to v)/k(v \to u)) = -\beta(E_v - E_u)$ against independently measured energy differences. Drive the residue and measure the current that circulates around the class triangles. Which constitutive law the world uses is an experimental question, and the topology has already said which numbers decide it.

11.7 The exchange in matter

Any network on which two indistinguishable particles hop realizes an arena. The prediction is classical and sharp: on a 3-connected non-planar network the exchange phase is ±1; on a planar one it is free [21]. The witness itself is a twelve-site design whose planar core is made non-planar by the same leaves that give its reader a direction, and the scaled family supplies larger designs whose free part is known in closed form, $12n^2 - 3n + 1$.

11.8 The integral side

Two integral data are now in hand that no rational reader can see: the order-3 clock character, which exists modulo d and not over $\mathbb{Q}$, and the order-2 exchange, which vanishes over $\mathbb{Q}$. On the witness the rational side of the relation-generation problem is closed — 15 orbit generators, attained [4] — and that says nothing about the integral side, where the relation gap problem and Wall's D(2) problem live. These two data are the first concrete footholds there.

11.9 Retained continuation as a theorem

When can a history be compressed into an organization that is sufficient for all future transitions, and what is the smallest such organization? For a fixed set of continuations and a fixed observation the answer

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is Keyframe 0 — the address, the class of histories that no allowed continuation separates — and its instances in this paper are proved: one bit for the turn, the reader cut for the adder, trace and swap for loops of two tokens. Its realization form is Keyframe 17: two systems that keep the same addresses play the same game. The open case is a system whose retained organization changes its own effective rule (§8.2), so that what can tell two histories apart is itself moved by what the system keeps. There the boundary between reacting and reorganizing can be measured, and a candidate law — the result of resolving a continuation is kept as organization that changes the accessibility of later continuations — can be tested against a system that could fail it.

11.10 The inverse problem

Given an observed sequence $a(n) = \pi(T^n(x_0))$, recover the smallest (X, T, π) that generates it. For π in base 16 the read-head is in hand (§4.6), and Keyframe 6a says when a digit converted to another base is resolved. For SHA-256 the honest state is also a keyframe. With its context supplied the function runs backward exactly — the schedule from any 16 consecutive words, each round from its word (Corpus result 3). Without that context, one reverse round is still a closed-form computation; the true message word scores exactly zero under a probe that reads only carry geometry; and when depth is added without an answer key, the measured collapse factor of the cold basin is 1.00 per round in the schedule-free model — depth tightens the basin precisely as much as the search for each new word widens it [14]. The difficulty is relocated, not destroyed, and we know where it lives: in the words that are not supplied. The message schedule, which couples them, is the named next experiment. The hexadecimal transformation space — how a single source expands under every way of reading it — is being mapped in parallel.

11.11 The standing four

standing aim	where it stands	keyframes
1. how a transformation is done, not its values	the address action; π read as an angle; alignment buys cost, not possibility; a converted digit is resolved by its boundary; the two seeds are powers of a pole	6, 6a; Corpus result 1
2. shapes are methods	literal: the start is a shape, the swap; the walker's interior is the Cayley graph of its methods; every path is continue or turn; the methods form a device for logic, and the device is the walk	1, 7–11, 17
3. read the missing methods without searching	an unknown is an address already fixed; the deficiency is a module, $Q = V/W$; no zero of the field; hidden channels at computable addresses; one character answers every channel; trace and swap classify every loop	0, 0b, 4–4c, 12
4. something real to shoot for	SHA-256 runs backward with its context, and the cold basin is balanced without it; exchange statistics on networks; the rate law's named unknowns; the integral side; Navier–Stokes and a natural branching system, such as protein folding, as targets with their missing theorems named	13–16a; §§10–11

Table 12. The programme's four standing aims, located.

12. Status ledger

	statement	status	source
0	the address is the object: a state is the class no allowed continuation separates; splitting by failures stops there	recovered; computed; proved	§3.1; [28, 35]; App. A.1
0a	one field, three readers: 24, 8 and 3 addresses; $X/\sim \cong \mathrm{im}\ \Phi$	computed; proved	§3.2; App. A.1

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	statement	status	source
0b	a reader is a projection; the bins 115 + 120 from the multiplicity vector; export at second order on the walk	computed; recovered; proved	§3.3; [27]; App. A.18
0c	transport has a baseline: $T_a \cap T_b = T_{\gcd(a,b)}$; residual lcm − a − b + gcd	proved; computed	§3.6; App. A.19
0d	reachability depends on the boundary alone, the address on boundary and reader together	computed	§3.7; Table 2
1	the only possible start: nothing on one state, only the swap on two; 5 and 83 above	computed, $n \le 4$; proved for one and two states; recovered	§2.1; [24]; App. A.17
1a	under both prohibitions every two-block reading contains both crossings of the swap; only the swap is read without rest	proved; computed	§2.2; App. A.17
1b	the swap has no fixed point, so no reader reads all of its own readings	recovered; computed; proved	§2.3; [40, 62]; App. A.17
2	C1 on the cycle ⟺ Abrams' hypothesis	computed; recovered	[15]; [19]
3	seen + hidden = $\log_2 N!$; orientation is a holonomy	proved	[9, 12]
4	one character answers every channel; the most symmetric reader is blindest	computed; recovered	[4]; [27]
4a	a reader is a functional; three kinds of zero; no zero of the field on the witness	proved; computed	[3]; [18, §§9–20, 34]
4b	the invariant read is bin 0; the rest at a computable address; export at second order	computed; recovered	[18, §§35–49]; [27]
4c	the vocabulary × reader interface has width 6, and less than either slot when both are restricted	computed; recovered	[18, §§24–29]; [39]
5	adder cut law; $b_1 = C(n, 2) + (n - 1)$; carry capacity	proved, all n; computed	[6–8]
6	reading at an address; alignment buys cost	measured earlier; digits re-run	[17]
6a	a converted digit is resolved by its boundary; bounded exactly for c = 2, 4, 8, 16	proved; computed	§4.6; App. A.6
7	the interior is the Cayley graph of the methods	computed	[5]
8	continue and turn are the only rules with one more step	computed	[18, §65]
9	one remembered bit is the minimal memory	proved	[18, §66]
10	the device; AND only where the reader is small	computed	[18, §67]
11	turn, clock and one interior at every size	proved, $n \ge 3$; computed n = 3, 4	[18, §§69–71]
12	trace and swap: $H_1 \cong H_1(\Gamma) \oplus \mathbb{Z}/2$	computed n = 2…6; every n via [21]	[18, §§74–75]
13	the exchange is the orientation character of the $K_{3,3}$ surface	computed	[18, §74]
14	the residue: $H^1 \cong \mathbb{Q}$ for every n	proved via [21]; computed	[18, §§72–75]
14a	the residual counts what the declared relations fail to span, at every order	proved; computed	[37]
15	the clock is the residue modulo 3	proved; computed $n \le 4$	[18, §§68–74]
16	prices and drives, 22 = 7 + 15, then 2, 1, 0; the exchange is phase-only	computed; proved	[18, Part I, §75]

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	statement	status	source
16a	the operator is not forced; a zero price is a move the reader cannot see	proved; computed	[18, §§50–54]
17	the field and the computer are one object when their addresses agree	computed; recovered	§8.5; [28, 35, 39]
—	the constitutive law: mobilities and scale	open; the deciding numbers named	§7.3

Table 13. Every keyframe, with its status and where it is on record.

13. What was corrected on the way

The programme keeps its killed claims as navigational markers; its audit papers record them in full. Twenty-one of them shaped the keyframes above — the last nine came with this version, from the corpus review and from an independent check of the new sections — and each was settled by running something, not by argument alone.

claim	what replaced it	how it was found
at $n = 3$ a graph gives turn or clock, not both	both, once the face marks have no centre; the failure was an artefact of a fixed centre	building the marks without centres
the trace map is an isomorphism	an exact sequence whose kernel is the exchange; an isomorphism over $\mathbb{Q}$	a review, re-run
the residue is a blind spot for odd cycles	omission from the declared family: arenas with no odd cycle carry it too	bipartite arenas of C_6 and C_8
an odd interior factor forbids a decomposition into digits	true in base 2 only: $b = 3$, $L = 2$ gives $9 = 3^2$	changing the base [7]
a walk's endpoint is its phase plus one parity	it needs the turn parity at each of three positions	all 8190 words
an unknown is a gap to be filled; an object occupies an address	the address is the object; an unknown is an address the relations already fix	Dean's correction, then run: splitting by failures on the walk, $6 \to 12 \to 24$
the readers that determine the module are the six cyclic classes	1 of the 357 minimal sets is cyclic; 356 use a non-cyclic class	enumerating all 4,095 subsets [18, §22]
nothing is destroyed by a read	restriction preserves the representation; keeping only the invariant projection discards the rest	a review of the bin decomposition [18, §42]
the gradient rule cuts the weights to a 7-dimensional subspace and excludes the Laplacian	7 is a count of parameters; the weights span 18; the Laplacian is excluded only if a zero price forces a zero weight	a review, run over 140 potentials [18, §52]
the price-blind faces are blind by a mismatch of degree	reader invariance: an exact price for the order-8 reader gives all 14 faces a nonzero length	re-run in the arena's own coordinates [18, §57.9]
a delayed positive loop has a stable cycle past its local threshold	a long transient: the oscillating fraction falls as the run lengthens	lengthening the run [36]
the deck group of the cover is inertia	the period group is $d\mathbb{Z}$; cost needs a functional; using a channel (3) and erasing it (4) differ	breadth-first search on the cover [15]
the start is unique however *cannot hold still* is read	unique when it is read as *must*; read as *can*, two states carry three structures, all with the swap's moves	running five readings (`start.py`)

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claim	what replaced it	how it was found
no terminal state and no fixed point are one condition	three conditions: no terminal state admits one state resting forever; not its own only continuation (reading D) kills one state but admits rests; only no fixed point keeps the start unique, and Lawvere's theorem needs it too	reading D against reading A; the corpus review [50, 63]
a rise in rank or dimension shows something new	novelty is a quotient over the declared transport; name the transported part first	the Move Layer's protocol [41]; Keyframe 0c
the SHA round and schedule are reversible, so the digest is invertible	reversible with the context supplied; the digest is a separate many-to-one projection	re-run with and without the round's word (Corpus result 3)
exactly two seeds across all kernel frames	two in the positive primitive secant class; the frame (−64, 12) lies outside it	the corpus's kernel-frame analysis [46]
the census of strongly connected digraphs differs from the start at one state by a loop	by a state with no continuation: the census has no loops, and its one-state digraph is a state with nowhere to go	an independent check of this version
what is allowed and what is read are separate axes	they are separate inputs, and the address depends on both: withdrawing one move takes a coarse reader on the walk from 6 addresses to 4	an independent check; the boundary-by-reader table (Table 2)
the interaction residual is decided by $R_{15} - 9$	by the quotient over the transported span; the difference alone nets new directions against collisions	an independent check of this version
the successor on the integers never freezes its chain of images	on the natural numbers it never freezes; on the integers it is onto, and its chain is frozen from the start	an independent check of this version

Table 14. Corrections that shaped the keyframes.

14. Method, authorship and reproducibility

The programme is driven by Dean A. Kulik. Computation, verification and drafting were carried out with Claude (Anthropic) in long working sessions. Drafts and results were also reviewed in parallel sessions with other AI systems, and every checkable point in those reviews was re-run before it was accepted or rejected; where an independent run reported the same numbers, that is noted. The wider corpus was reviewed as a whole by one such review of forty papers and a revised synthesis of it [63]; its checkable items were re-run here, and the rest are marked reported. The new sections of this version were then checked by a separate pass that had not seen them written; every finding was re-run, and the corrections are in Table 14.

All computations are exact — integer, rational or finite-field arithmetic, with one scan at 80 significant digits whose hits are then checked exactly — and use the Python standard library, with NetworkX for planarity tests and SymPy for symbolic work. Every number in this paper is drawn from the programme's ledger [18], where each result is recorded with the script that produced it, its controls and its scope. The scripts for the results first reported here are `start.py`, `math_core.py`, `corpus_rerun.py`, `n4.py`, `n56.py`, `surface.py`, `graph4.py`, `kpoint.py`, `planar_contrast.py`, `location.py`, `realize.py` and `keyframes_check.py`. All twelve were re-run for this version; Appendix B lists each with its hash, its running time and what it printed, and Appendix A gives the proof of every statement marked proved. Results marked measured earlier come from working sessions recorded in the ledger, some of them

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without an archived repository, and the ledger says which [18, §48]. This paper is released under CC BY-NC 4.0.

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Appendix A. Proofs

Every statement marked *proved* in the paper is proved here, or in the source cited beside it; Keyframe 5 is proved in [6] and [7], and the results recovered from the literature are used with their hypotheses checked. Statements marked *computed* are exact finite computations, and Appendix B lists their scripts.

A.1 Keyframes 0 and 0a, and the squares of §3.5 and §8.5. Let a system have configurations X, actions acting by $x \mapsto x{\cdot}a$, and an observation O on X. Put $x \sim y$ when $O(x{\cdot}w) = O(y{\cdot}w)$ for every word w of actions. (i) The relation is a right congruence: if $x \sim y$ then $x{\cdot}a \sim y{\cdot}a$, since every word after a is a word aw. So $T_a[x] = [x{\cdot}a]$ is well defined, the class map q satisfies $q(x{\cdot}a) = T_a(q(x))$, and O is constant on classes (take w empty). (ii) Let r be any reading such that $r(x{\cdot}a)$ is fixed by $r(x)$ and a, and $O(x)$ by $r(x)$. If $r(x) = r(y)$, then $r(x{\cdot}w) = r(y{\cdot}w)$ for every w by induction on its length, so $x \sim y$: every sufficient organization refines $\sim$, and the classes are the smallest one, unique up to renaming. (iii) If r merges x and y with x and y in different classes, some word w gives $O(x{\cdot}w) \neq O(y{\cdot}w)$. Moore's refinement — partition by O, then repeatedly by the class of x together with the class of $x{\cdot}a$ for each action a — splits along exactly such words and stops at $\sim$, after at most as many rounds as there are classes [28, 35].

A.2 Keyframe 17 and §3.5. Let (X, F, O) and (Y, F', O') be realizations with the same actions and observation values, and $q: X \to Y$ onto with $q(F_a x) = F'_a(q\,x)$ and $O'(q\,x) = O(x)$. Then $O(F_w x) = O'(F'_w q\,x)$ for every word w, so $x \sim x'$ exactly when $q\,x \sim q\,x'$, and q induces a bijection of classes that commutes with every action and keeps the observation: the two smallest machines are one machine. For the walk and the register, q is a bijection of 24 states, the square holds on all 48 pairs of state and action — which is the

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whole statement, since every word is a composite of single actions — and the current vertex is read from the phase and the newest bit without error.

A.3 Keyframe 2. A configuration of K tokens on C_n is stuck exactly when every token has both neighbours occupied. If $K < n$ there is an empty vertex, and walking round the cycle from it one reaches a token beside an empty vertex, which can move; if $K = n$, nothing moves. Abrams' condition on C_n asks that the one essential loop have at least $K + 1$ edges, $n \geq K + 1$ [19, 20]: the same inequality.

A.4 Keyframe 3. A reader that sees an observation o of a uniformly random permutation of N items receives $\log_2(N!/|R^{-1}(o)|)$ bits, and the fibre $R^{-1}(o)$ holds $\log_2|R^{-1}(o)|$ more; the two add to $\log_2 N!$ for every o [9].

A.5 Keyframe 4a. Over $\mathbb{Q}$, averaging over H projects onto H-invariants, so a family read at H spans the H-invariant part of what it spans at the trivial reader. Taking invariants is exact when $|H|$ is invertible (Maschke), so for every quotient $Q = W_R/(W_R \cap W_S)$, $\dim Q^H = \dim W_R{}^H - \dim (W_R \cap W_S)^H$, which is the loss measured at H. Decomposing a module V into irreducibles gives $\dim V^H = \Sigma_\lambda\, m_\lambda(V)\, \dim \lambda^H$: a multiplicity vector that does not depend on the reader, paired with the reader's fixed-vector counts. If $H \leq K$ then $V^K \subseteq V^H$, so every count is monotone in the reader. For a cyclic reader the fixed-vector count is the zero-frequency coefficient of the discrete Fourier transform of the character on the orbit, and the coefficients sum to the dimension. Faithful-reader lemma: if $U \subseteq V$ and $\dim U^H = \dim V^H$ at a reader with $\dim \lambda^H > 0$ for every λ, then $\Sigma_\lambda\, (m_\lambda(V) - m_\lambda(U))\, \dim \lambda^H = 0$ with every term non-negative and every weight positive, so $m(U) = m(V)$ and $U = V$, and the equality holds at every reader [3; 18, §§8–16a].

A.6 Keyframe 6a. Let $x \in [0, 1)$ be read in hexadecimal. After k digits x lies in a cell $[m/16^k, (m + 1)/16^k)$, and $\lfloor cx \rfloor$ is constant on that cell unless a boundary d/c, $1 \leq d < c$, lies strictly inside it. If d/c has a terminating hexadecimal expansion of length k_0, then for $k \geq k_0$ it is a cell end and never strictly inside. If it does not terminate, it lies strictly inside the cell that contains it at every k, and the input $x = d/c$ is never resolved. So the worst case is bounded exactly when every d/c terminates — when c divides a power of 16, that is $c = 2$, 4, 8 or 16 — and for any c at most $c - 1$ cells are unresolved at each k, a share of at most $(c - 1)/16^k$.

A.7 Keyframe 8. Continue sets the next vertex to pp and turn to its antipode, so every step gives $v[k + 3] = \varepsilon_k v[k]$ with $\varepsilon = \pm 1$. Hence $v[3j + r]$ is $v[r]$ times the product of the ε at the positions congruent to r modulo 3. The last three vertices of a walk are therefore fixed by its length modulo 3, which fixes their axes, and by the parity of the turns at each of the three positions, which fixes their signs: $3 \times 8 = 24$ endpoints, all distinct.

A.8 Keyframe 9. If a reader element g fixes a state s, an equivariant rule T satisfies $T(s) = T(gs) = g\, T(s)$, so $T(s)$ must be a continuation that g fixes. On the witness every non-identity element that fixes an interior state exchanges its two continuations, so no rule exists. A memory that admits a rule must therefore have no point fixed by a non-identity element, so each of its reader orbits is regular, of size $24 = 2 \times 12$ — one bit more than the 12 states. With trivial stabilisers a rule does exist: choose a continuation at one state of each orbit and carry it round the orbit [5; 18, §66].

A.9 The curvature of the method surface (§5.3). On a closed triangulated core with N triples, the method surface has N vertices and $2N$ edges, one continue and one turn leaving each triple. Continue closes in triangles, $N/3$ faces; turn closes in its turn loops; the remaining faces are squares using two continue and two turn edges each, $N/2$ of them, since every edge lies on two faces. So $\chi = N - 2N + N/3 + N/2 + (\text{turn loops}) = (\text{turn loops}) - N/6$.

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A.10 Keyframe 11. A face automorphism of the sum-rule triangulation of $K_{n,n,n}$ that fixes each class has the form $(x, y, z) \mapsto (\lambda x + a, \lambda y + b, \lambda z + c)$ with $\lambda = \pm 1$, and $a + b + c \equiv 0$ if $\lambda = 1$, $\equiv 1$ if $\lambda = -1$: faces that share an edge lie in different families and the dual graph is connected, so the two families are kept or exchanged globally, and the face equations force all three coordinate maps to be affine with one common unit slope. That gives $2n^2$ class-fixing automorphisms, and with the three class rotations $|G_n| = 6n^2$. Degrees separate core, leaves and marks; the marks follow their corners and the leaves keep the cyclic order of the classes, so the automorphism group of the marked graph is G_n. Translations are transitive on the $3n^2$ core edges, and no class rotation fixes a vertex, so none fixes a pair of token positions [18, §§70–71].

A.11 Keyframe 12. (i) The swap ε vanishes on every commuting square, since two tokens that move on disjoint edges and back do not exchange; the trace J vanishes on it too, each edge being crossed once each way. Both are therefore defined on H_1. (ii) The exchange z traces nothing and swaps, so $z \neq 0$, and a 2-chain B of commuting squares has $\partial B = 2z$ (22 squares on the witness, 18 on two classes of the core for $n \geq 3$), so z has order 2. (iii) For every cycle c of the graph, one token going round c while the other waits off it is a legal loop with $J = c$ and $\varepsilon = 0$, so J is onto $H_1(\Gamma_n)$. (iv) By Theorem 4 of [21], a simple 3-connected non-planar graph has $H_1(D_2\Gamma) \cong \mathbb{Z}^{\beta_1} \oplus \mathbb{Z}/2$. Γ_n is simple; it contains $K_{3,3}$; and removing any two vertices leaves the core connected, since $K_{n,n,n}$ is $2n$-connected, with each leaf still joined to one of its $n + 1$ core neighbours and each mark to one of three disjoint routes. Since J maps onto a free group the sequence splits, so the kernel of J is a group of order 2 containing z, that is $\{0, z\}$; and ε separates its two elements. So (J, ε) is an isomorphism, and every Y-exchange, which traces nothing and swaps, is z.

A.12 Keyframe 14. (i) Over $\mathbb{Q}$, J is an isomorphism (A.11). (ii) A closed walk of four moves traces a closed chain of at most four edge traversals. If the chain is not zero it is a 4-cycle of the graph traversed once: a triangle would leave one traversal unmatched, and the graph has no multiple edges. If it is zero, two edges are each crossed out and back; the same edge, or two edges sharing a vertex, cannot be crossed that way by two tokens that never collide without repeating a configuration, so the edges are disjoint and the walk is a commuting square. Every 4-cycle of the graph is traced by a spectator square, since a waiting vertex off it exists. So filling every arena 4-cycle fills, through J, exactly the graph's 4-cycles. (iii) In $K_{n,n,n}$ a cycle of length at least 5 has a chord, so modulo 4-cycles every cycle is a sum of triangles, and two triangles sharing an edge differ by a 4-cycle; a path through a leaf is a core path modulo one 4-cycle; and a mark edge is the path through its corners modulo its prism square. So $H_1(\text{graph} + 4\text{-cycles}; \mathbb{Q})$ is spanned by one triangle, and the cocycle f, ± 3 on every triangle, shows it is not zero [18, §§74–75].

A.13 Keyframe 15. Edge by edge $f \equiv \text{cl}(v) - \text{cl}(u)$ modulo 3: a forward core edge carries 1, a leaf to its own core point $3 \equiv 0$, a leaf to the next class 1, a mark to its corner 0, and a mark edge the value of its corners' edge. So modulo 3 the cocycle is exact with potential cl, and on the arena with potential $\text{cl}\, u + \text{cl}\, v$, whose negative is $F = 2(\text{cl}\, u + \text{cl}\, v)$. A class rotation g adds 1 to every class, so $F(gx) = F(x) - 2 \equiv F(x) + 1$: F is single-valued and shifted by the character rot. If a class rotation fixed a configuration x, the shift would contradict $F(gx) = F(x)$, so the clock survives exactly when no class rotation fixes a pair of token positions, which holds whenever none fixes a vertex [18, §§68–69].

A.14 Keyframe 14a. At any order, a declared cell is a cycle, and the differential evaluates a cochain on the declared cells, so its kernel is the annihilator of their span. Over a field the exact cochains are the annihilator of the whole cycle space Z at that order, since both have dimension $\dim C - \dim Z$. Subtracting, $\dim \ker - \dim \text{im} = (\dim C - \dim \text{span}\, \mathcal{K}) - (\dim C - \dim Z) = \dim Z - \dim \text{span}\, \mathcal{K}$. The same subtraction holds in the invariant sector, with invariant subspaces throughout [37].

A.15 Keyframe 16. B is a combination of commuting squares with $\partial B = 2z$ (A.11). If the antisymmetric part a has zero affinity round every commuting square, $\oint_{\partial B} a$ is a signed sum of square affinities and

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vanishes, so $\oint_z a = \tfrac{1}{2} \oint_{\partial B} a = 0$. A real rate law can put no circulation on the exchange; a phase can, through the characters of $\mathbb{Z}/2$.

A.16 Keyframe 16a. Let p be a G-invariant price with $p = d\varphi$ on the connected arena. For $g \in G$, $d(\varphi \circ g - \varphi) = g \cdot p - p = 0$, so $\varphi \circ g - \varphi$ is a constant $c(g)$; then $c(gh) = c(g) + c(h)$, a homomorphism from a finite group into $\mathbb{Q}$, so $c = 0$ and φ is invariant. Hence $p(u \to v) = \varphi(v) - \varphi(u) = 0$ whenever u and v lie in one orbit — on the witness, the 24 moves inside one class of configurations — and if φ takes a different value on each of the eight classes these are its only zeros [18, §§51–54].

A.17 Keyframes 1, 1a and 1b. (i) On one state a the only possible continuation is $a \to a$. The strict readings forbid it, and without it a is terminal; reading D asks for a continuation other than a, and there is none. So no reading admits one state. (ii) On two states each state needs a continuation other than itself, and the only other state is the other one, so $a \to b$ and $b \to a$ are forced. With rests forbidden the relation is exactly the swap, which is strongly connected, a map with one orbit and a bijection; so readings A, A0, B and C each admit it and nothing else. With rests allowed (reading D) each of the two rests is optional: four labelled structures, three up to exchanging a and b. (iii) Let G obey both prohibitions — reading A, B or D — on $n \geq 2$ states, and let a reader split the states into nonempty blocks B_0 and B_1. Take $x \in B_0$ and $y \in B_1$. A path from x to y exists by strong connectivity, and its first step out of B_0 is an edge from B_0 to B_1; a path from y to x likewise gives an edge from B_1 to B_0. If G is in reading A and $n \geq 3$, take any edge $u \to v$: the split of $\{u, v\}$ against the rest, which is nonempty, reads that edge as a rest. If $n = 2$ the only split separates the two states, and the swap has no edge inside a block. A cycle through all n states is read without a rest exactly when consecutive states lie in different blocks, that is, when the blocks alternate round the cycle, which is possible exactly when n is even. (iv) For any $f\colon A \to 2^A$ put $d(a) = s(f(a)(a))$. Since s has no fixed point, $d(a) \neq f(a)(a)$, so $d \neq f(a)$ for every a. In Lawvere's form [40]: if $\varphi\colon A \to Y^A$ is onto, then for every $t\colon Y \to Y$ the map $g(a) = t(\varphi(a)(a))$ equals $\varphi(a_0)$ for some a_0, and evaluating at a_0 gives $t(\varphi(a_0)(a_0)) = \varphi(a_0)(a_0)$, a fixed point of t. The swap has none.

A.18 Keyframe 0b. Let ρ be a representation of a finite group H on V over $\mathbb{Q}$ and $R = (1/|H|)\ \Sigma_h\ \rho(h)$. For $g \in H$, $\rho(g)R = R$, since multiplying by $\rho(g)$ permutes the terms; so the image of R is invariant, and $R^2 = (1/|H|)\ \Sigma_g\ \rho(g)R = R$. An invariant vector is fixed by every term, so $Rv = v$, and the image of R is exactly V^H. The rank of an idempotent is its trace, which gives $\dim V^H = (1/|H|)\ \Sigma_h\ \mathrm{tr}\ \rho(h)$. For a central involution z, $P_\pm = (1 \pm \rho(z))/2$ are complementary idempotents. Since z is central it acts on each rational irreducible type through that type's endomorphism ring, a division algebra (Schur), where $z^2 = 1$ leaves only $+1$ and -1; so on each type one of $P_\pm$ is the identity and the other is zero, and $\dim P_\pm V = \Sigma\ m_\lambda \dim \lambda$ over the types of that sign. An operator that commutes with $\rho(z)$ preserves both images, so linear dynamics that respect the symmetry never move a field from one bin to the other. In the representation on functions, $(fg)(zx) = f(zx)\ g(zx) = (-f(x))(-g(x)) = f(x)\ g(x)$ when f and g are z-odd, so the square of a hidden field is z-even and visible; it determines $|f|$ and not the sign of f.

A.19 Keyframe 0c. Let a and b divide N, and let T_a be the functions on $\mathbb{Z}/N$ with $f(x + a) = f(x)$. Its dimension is a, the indicators of the residue classes modulo a being a basis. A function with periods a and b has every period $ua + vb$, in particular $\gcd(a, b)$ by Bézout; conversely a function of period $\gcd(a, b)$ has periods a and b. So $T_a \cap T_b = T_{\gcd(a,b)}$, and $\dim(T_a + T_b) = a + b - \gcd(a, b)$. Both summands have period $\mathrm{lcm}(a, b)$, so $T_a + T_b \subseteq T_{\mathrm{lcm}(a,b)}$, and the quotient has dimension $\mathrm{lcm}(a, b) - a - b + \gcd(a, b)$.

A.20 Corpus result 1. Put $\theta = 2\pi/m$ and $c = 2\cos\theta$. Then $1 + i\tan\theta = \sec\theta\ (\cos\theta + i\sin\theta)$, so $(1 + i\tan\theta)^m = \sec^m\theta \cdot e^{2\pi i} = \sec^m\theta$. For $m \geq 3$, c is an algebraic integer of degree $\varphi(m)/2$ whose conjugates are the distinct real numbers $2\cos(2\pi j/m)$ with $\gcd(j, m) = 1$. If $\sec^m\theta = (2/c)^m$ is rational, so is c^m, and every conjugate $\sigma(c)$ satisfies $\sigma(c)^m = c^m$, hence $|\sigma(c)| = |c|$. Distinct real numbers with one absolute value

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number at most two, so $\varphi(m) \le 4$ and $m \in \{3, 4, 5, 6, 8, 10, 12\}$. For $m = 5$ and 10 the two conjugates, $(-1 \pm \sqrt{5})/2$ and $(1 \pm \sqrt{5})/2$, differ in absolute value, and for $m = 4$, $c = 0$. The remaining cases, with $m = 1$ and 2, give $\sec^m \theta = 1, 1, -8, 64, 16$ and $4096/729$ for $m = 1, 2, 3, 6, 8$ and 12. The integers greater than 1 are 64, at $m = 6$, and 16, at $m = 8$.

A.21 Corpus results 2 and 3. For an integer $S \ge 0$, division with remainder gives unique W and Q with $S = W + 2^{32}Q$ and $0 \le W < 2^{32}$, so (W, Q) determines S and S determines (W, Q); for a sum of two 32-bit words $Q \in \{0, 1\}$, and S and $S + 2^{32}$ have the same W whenever both occur, so W alone is two-to-one at every value except the top one, which has a single preimage. For SHA-256, the round sends $(a, \ldots, h)$ to $(T_1 + T_2, a, b, c, d + T_1, e, f, g)$ with $T_1 = h + \Sigma_1(e) + \mathrm{Ch}(e, f, g) + K_t + W_t$ and $T_2 = \Sigma_0(a) + \mathrm{Maj}(a, b, c)$ modulo 2^{32} [64]. The new state gives a, b, c, e, f, g directly, hence T_2, hence T_1 from the new a, hence d from the new e; and $h = T_1 - \Sigma_1(e) - \mathrm{Ch}(e, f, g) - K_t - W_t$ is a translate of $-W_t$, one-to-one in it. The schedule $W_t = \sigma_1(W_{t-2}) + W_{t-7} + \sigma_0(W_{t-15}) + W_{t-16}$ for $16 \le t < 64$ solves for W_{t-16}; given $W_k, \ldots, W_{k+15}$, applying it at $t = k + 15, k + 14, \ldots$ recovers the words below k one at a time.

Appendix B. Scripts, hashes and what they printed

All twelve scripts were re-run for this version, in Python 3.11.15 with NetworkX 3.6.1, SymPy 1.14.0 and mpmath 1.3.0, in exact arithmetic, apart from the 80-digit scan for the seeds, and without randomness. The homology scripts build every graph and arena themselves; files uploaded from other sessions are read only to cross-check their numbers. Times are wall-clock on one core.

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script	what it computes	what it printed in this run	time
`start.py`	the start: five readings for $n \le 4$; the sixteen relations; two-block readers; the diagonal	0, 1, 5, 83 in reading A; the swap alone on two states in A, A0, B, C, three in D; both crossings in every structure of readings A, B and D, none or one without *cannot escape*; the diagonal outside the image for all 2, 16 and 512 maps	3 s
`math_core.py`	three readers on the walk; averaging; bins from multiplicities; transfer; periodic transport; two waves	24, 8 and 3 addresses; ranks 12 and 8; 115 + 120, 105 + 111, 10 + 10; all 144 divisor pairs; 4,368 echoes; every entry of Table 2	4 s
`corpus_rerun.py`	the seeds; the carry; SHA-256 with its context	integers greater than 1 only at $m = 6$ and 8; the 4-bit fibres; 2 of 2 digests, 49 of 49 windows, 64 of 64 rounds	1 s
`n4.py`	two tokens on Γ_2, Γ_3, Γ_4: integral H_1, the exchange, the trace map	$\mathbb{Z}^{19} \oplus \mathbb{Z}/2$, $\mathbb{Z}^{100} \oplus \mathbb{Z}/2$, $\mathbb{Z}^{181} \oplus \mathbb{Z}/2$; $\partial B = 2z$; all 432, 10,260 and 32,352 Y-exchanges equal z	3 s
`n56.py`	Γ_5, Γ_6: integral H_1; trace and swap	$\mathbb{Z}^{286} \oplus \mathbb{Z}/2$, $\mathbb{Z}^{415} \oplus \mathbb{Z}/2$; $\varepsilon = 0$ on all 102,330 and 212,049 squares; parked lifts 286/286 and 415/415	12 s
`surface.py`	the two-token surface of $K_{3,3}$	15 / 36 / 18, $\chi = -3$, non-orientable, genus 5; cup form of rank 5; $w_1(z) = 1$	1 s
`graph4.py`	the graph with its 4-cycles filled, n = 2, …, 8; arena 4-cycles by trace	dimension 1 for n = 2, …, 8, $f \equiv$ d(cl) mod 3; at n = 2, 528 + 264 + 291 arena 4-cycles	13 s
`kpoint.py`	k = 2, 3, 4 tokens, subdivided and raw	every entry of Table 8	5 s
`planar_contrast.py`	nine 3-connected graphs	every entry of Table 7	< 1 s
`location.py`	classes of sums; refinement on the walk; the serial adder	28 classes; 6 → 12 → 24; 5 + 3 without its carry reads 6	< 1 s
`realize.py`	field and computer (Keyframe 17); digit resolution (Keyframe 6a)	48 of 48 pairs; 2,046 words agree; bounded exactly for c = 2, 4, 8, 16	< 1 s
`keyframes_check.py`	shapes; the BBP integral; hex digits; Kronecker; Dedekind	0, 1, 5, 83; π; 243F6A8885A3; $x \cdot x = 2$; 1.414213562373095	1 s

Table 15. The scripts of this paper. First sixteen hexadecimal digits of each file's SHA-256: start.py ad1c99ddcbcca61d, math_core.py 2e73103b443c0144, corpus_rerun.py be1b9d054fbec9e4, n4.py 01d678c22ed096d0, n56.py 21529f94c0e082f4, surface.py 86060c23c7bea159, graph4.py a1e949105970717a, kpoint.py 7dcb6c72fcc52364, planar_contrast.py 9466eba0376465f8, location.py 8065ba96ca1c1bd4, realize.py 7eeba3606be22ed2, keyframes_check.py cd976602b448b697.

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